

Large Sample Approximation for Least Squares Support Vector Regression

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Editor: My editor

Abstract

This paper examines the asymptotic performance of Least Squares Support Vector Regression (LS-SVR) in a large-scale high-dimensional context, where both the number of observations n and the number of features p increase to infinity, maintaining a fixed ratio $p/n \rightarrow \bar{c}_0 \in (0, +\infty)$. Unlike previous studies that primarily focused on classification with Gaussian-distributed covariates, we extend the theoretical framework to the regression problem and analyze the effects of sub-Gaussian input distributions on the LS-SVR model. Using kernel approximations and asymptotic analysis, we derive closed-form expressions for the regression function and provide insights into LS-SVR's behavior under high-dimensional scaling. We theoretically demonstrate that, under certain regularity conditions, the LS-SVR approximation converges almost surely to the full LS-SVR solution and establish the asymptotic normality of LS-SVR. Simulation results support theoretical approximations and reveal how key parameters, such as kernel bandwidth and regularization constant, influence model performance.

Keywords: Least Squares Support Vector Machines (LS-SVM), Least Squares Support Vector Regression (LS-SVR), High-Dimensional Asymptotics, sub-Gaussian Distributions, Large-Sample Approximation

1. Introduction

Support Vector Machines (SVM), introduced by Cortes and Vapnik (1995) for binary classification with nonlinear decision boundaries, is a popular supervised learning algorithm.

In classification tasks, SVM aims to identify the hyperplane that best separates different classes in the feature space. SVM maximizes the margin to create a robust classification boundary that is less susceptible to noise and outliers. SVMs have proven effective across a range of applications, including image classification, text categorization, and bioinformatics. It can also be used for regression tasks, called Support Vector Regression (SVR) (Vapnik, 1999). SVR models complex non-linear relationships using an ϵ -insensitive loss function, which allows for some prediction error (Drucker et al., 1996). Due to its robustness and ability to handle large datasets (Smola and Schölkopf, 2004), SVR has proven effective across fields such as finance, engineering, and bioinformatics.

Since the introduction of SVMs, there has been growing interest in improving their efficiency and scalability for large, complex datasets. This has led to the development of various extensions, including more sophisticated versions of SVM (Schölkopf and Smola, 2002), multi-class SVM (Shawe-Taylor and Cristianini, 2004), least squares SVM (Suykens and Vandewalle, 1999), and fixed-size LS-SVM (De Brabanter et al., 2010). Least Squares SVM (LS-SVM) is a computationally efficient variant of the original SVM. Using a different objective function, LS-SVM transforms the problem into a linear system of equations that can be solved efficiently, making it well-suited for large datasets.

The connections between least squares support vector machines (LS-SVM) and traditional support vector machines (SVM) were discussed in Ye and Xiong (2007). Regarding LS-SVM, a myriad of research exists, including a probabilistic interpretation of LS-SVM based on a Bayesian inference approach (Van Gestel et al., 2002), a comparison between LS-SVM and extreme learning machines (Huang et al., 2011), and an analysis of the asymptotic performance of LS-SVM as the sample size approaches infinity ($n \rightarrow \infty$) to determine optimal convergence rates. The connections between least squares support vector machines (LS-SVMs) and traditional support vector machines (SVMs) were explored in Ye and Xiong (2007). There is a wealth of research on LS-SVM, including a probabilistic interpretation based on a Bayesian inference approach (Van Gestel et al., 2002), a comparison between LS-SVM and extreme learning machines (Huang et al., 2011), and an analysis of the asymptotic performance of LS-SVM as the sample size approaches infinity, as well as optimal convergence rates (Caponnetto and Vito, 2007; Steinwart et al., 2009).

Kammoun and Alouini (2021) investigated the asymptotic behavior of the soft-margin and hard-margin SVM classifiers for $n, p \rightarrow \infty$ with $p/n \rightarrow \bar{c}_0 \in (0, \infty)$, where input data were drawn from a Gaussian mixture distribution. Liao and Couillet (2019) provided a performance analysis of LS-SVM for $n, p \rightarrow \infty$ and $p/n \rightarrow \bar{c}_0 \in (0, \infty)$. They considered a two-class classification model, assuming the input data follows a Gaussian mixture model with distinct mean vectors and covariance matrices for each class. They argued that, despite the clear non-Gaussianity of practical datasets, the behavior of some machine learning methods was similar under both Gaussian and deterministic input conditions. Kernel-based transfer learning methods under a k -class Gaussian mixture model for large-dimensional input data were studied in Ren et al. (2023).

While Liao and Couillet (2019) studied the asymptotic performance of the two-class classification LS-SVM model under the conditions where both the sample size n and dimensionality p approach infinity and the ratio $p/n \rightarrow \bar{c}_0 \in (0, \infty)$, there is currently no theoretical justification for how this asymptotic behavior applies to the least squares support vector regression (LS-SVR) model.

Next, we define the notation used throughout the paper. As $n, p \rightarrow \infty$, we will use the notation $O_p(\cdot)$ introduced in Couillet and Benaych-Georges (2016) which is borrowed from Benaych-Georges and Knowles (2018). It is defined for a random variable $X \equiv X_n$ and $u_n \geq 0$, if for any $\eta > 0$ and $D > 0$ we write $X = O_p(u_n)$ where $n^D \mathbb{P}(X_n \geq n^\eta u_n) \rightarrow 0$. Under Assumption 1, $x = O_p(u_n)$ is equivalent to either $O_p(u_n)$ or $O_p(u_p)$ because n and p scale linearly. For multidimensional objects, $\mathbf{v} = O_p(u_n)$ means the maximum entry of a vector (or a diagonal matrix) $\mathbf{v}$ in absolute value is of order $O_p(u_n)$ and $\mathbf{M} = O_p(u_n)$ means that the operator norm of $\mathbf{M}$ is of order $O_p(u_n)$. For matrix norms, we consider $\|\mathbf{M}\| = \sigma_{\max}(\mathbf{M})$ (largest singular value) and $\|\mathbf{M}\|_F = \sqrt{\text{tr}(\mathbf{M}^\top \mathbf{M})}$ which denote the operator and Frobenius norm of the matrix $\mathbf{M}$, respectively. Finally, we use the notations $\xrightarrow{d}$ and $\xrightarrow{a.s.}$ for convergence in distribution and almost sure convergence, respectively.

In this article, we describe the asymptotic performance analysis for the LS-SVR under the $n, p \rightarrow \infty$ and $p/n \rightarrow \bar{c}_0 \in (0, \infty)$ regime, where we consider the input data to be coming from a sub-Gaussian distribution and, as such, moving beyond earlier studies. Many distributions, such as Gaussian, Bernoulli, uniform, and any bounded zero-mean distributions, are sub-Gaussian (Rigollet and Hütter, 2015).

This paper is organized as follows. Section 2 briefly describes the LS-SVR model, and Section 3 introduces asymptotic approximation estimation and asymptotic distribution results, with all proofs provided in the Appendix. Section 4 provides simulation results of the proposed approximation, while Section 5 concludes by summarizing the main results and suggesting directions for future research.

2. Least Squares Support Vector Machines for Regression

Consider a training set of size n , a bivariate set $\{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)\}$ where $\mathbf{X}_i \in \mathbb{R}^p$. $Y_i \in \mathbb{R}$. The primal space formulation of the LS-SVR model is (Suykens et al., 2002)

$$\min_{\mathbf{w}, b, e} \mathcal{J}(\mathbf{w}, e) = \frac{1}{2} \mathbf{w}^\top \mathbf{w} + \frac{\gamma}{2n} \sum_{i=1}^n e_i^2 \quad \text{s.t.} \quad Y_i = \mathbf{w}^\top \varphi(\mathbf{X}_i) + b + e_i \quad (2.1)$$

where $i = 1, \dots, n$, $e_i \in \mathbb{R}$ are i.i.d. random errors with $\mathbb{E}[e|\mathbf{X}] = 0$, $\text{Var}[e|\mathbf{X}] = \sigma^2 < \infty$, $\mathbf{X} \in \mathbb{R}^{n \times p}$, and $\mathbb{E}[\mathbf{Y}|\mathbf{X}] = \mathbf{w}^\top \varphi(\mathbf{X}) + b$, $\mathbf{Y} \in \mathbb{R}^n$, $\varphi: \mathbb{R}^d \rightarrow \mathbb{R}^{n_h}$ is the feature map to the high-dimensional feature space (n_h can be infinite dimensional) as in the standard SVM case with $\mathbf{w} \in \mathbb{R}^{n_h}$, $b \in \mathbb{R}$, and $\gamma \in \mathbb{R}_0^+$ is the regularization parameter.

Using Lagrange multipliers with dual variable $\boldsymbol{\alpha} \in \mathbb{R}^n$ and $\mathbf{w} = \boldsymbol{\alpha}^\top \varphi(\mathbf{X}_i)$, KKT (Karush-Kuhn-Tucker) conditions can be used to obtain the solution of (2.1). The resulting linear system of equations is given by

$$\begin{bmatrix} 0 & \mathbf{1}_n^\top \\ \mathbf{1}_n & \boldsymbol{\Omega} + \frac{n}{\gamma} \mathbf{I}_n \end{bmatrix} \begin{bmatrix} b \\ \boldsymbol{\alpha} \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{Y} \end{bmatrix} \quad (2.2)$$

where $\mathbf{1}_n = (1, \dots, 1)^\top$ is a vector of ones with length n . By Mercer's theorem (Mercer, 1909), $\Omega_{ij} = \varphi(\mathbf{X}_i)^\top \varphi(\mathbf{X}_j) = K(\mathbf{X}_i, \mathbf{X}_j)$ for $i, j = 1, \dots, n$ with $K(\cdot, \cdot)$ a positive definite kernel function, $\mathbf{I}_n$ is an identity matrix of size n , $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)^\top$ and $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$.

The final LS-SVR model for an arbitrary point $\mathbf{x}$ is

$$\hat{g}(\mathbf{x}) = \sum_{i=1}^n \hat{\alpha}_i K(\mathbf{X}_i, \mathbf{x}) + \hat{b} \quad (2.3)$$

where $\hat{\alpha}$ and $\hat{b}$ are given by solving the linear system (2.2) exactly (De Brabanter et al., 2011)

$$\hat{b} = \frac{\mathbf{1}_n^\top \mathbf{S}^{-1} \mathbf{Y}}{\mathbf{1}_n^\top \mathbf{S}^{-1} \mathbf{1}_n} \quad \text{and} \quad \hat{\alpha} = \mathbf{S}^{-1} (\mathbf{Y} - \mathbf{1}_n \hat{b}) = \mathbf{S}^{-1} \left(\mathbf{I}_n - \frac{\mathbf{1}_n \mathbf{1}_n^\top \mathbf{S}^{-1}}{\mathbf{1}_n^\top \mathbf{S}^{-1} \mathbf{1}_n} \right) \mathbf{Y},$$

with $\mathbf{S} = \mathbf{\Omega} + \frac{n}{\gamma} \mathbf{I}_n$. In the remainder, we define the kernel matrix $\mathbf{\Omega}$

$$\mathbf{\Omega} = \left\{ f \left(\frac{\|\mathbf{X}_i - \mathbf{X}_j\|^2}{p} \right) \right\}_{i,j=1}^n \quad (2.4)$$

and the $n \times 1$ kernel vector in a point $\mathbf{X}$

$$k(\mathbf{X}) = \left\{ f \left(\frac{\|\mathbf{X} - \mathbf{X}_j\|^2}{p} \right) \right\}_{j=1}^n, \quad (2.5)$$

with $f(\cdot)$ a positive definite kernel function, e.g., the Gaussian radial basis function (RBF) with bandwidth $h > 0$, i.e.

$$K(\mathbf{X}_i, \mathbf{X}_j) = \exp \left(- \frac{\|\mathbf{X}_i - \mathbf{X}_j\|^2}{2ph^2} \right).$$

Next, we derive the asymptotic behavior of LS-SVR regression for large n and p . Specifically, we will discuss the large-sample and dimension approximation under the condition $p/n \rightarrow \bar{c}_0 \in (0, \infty)$ for $\hat{\alpha}$, $\hat{b}$, and the regression function given in (2.3).

3. Large Scale Approximation

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \in \mathbb{R}^p$ be independent vectors with unknown distribution F . Assume

$$\mathbf{X}_i = \boldsymbol{\mu} + \sqrt{p} \boldsymbol{\epsilon}_i \quad (3.1)$$

for some $\boldsymbol{\mu} \in \mathbb{R}^p$ and $\boldsymbol{\epsilon}_i$ follows sub-Gaussian distribution with variance proxy $p^{-1} \boldsymbol{\Sigma}$ and $\mathbb{E}[\boldsymbol{\epsilon}_i] = 0$ such that $\mathbb{E}[\mathbf{X}_i] = \boldsymbol{\mu}$. In the remainder of the paper, we will denote this as $\boldsymbol{\epsilon}_i \stackrel{\text{ind}}{\sim} \text{subG}(p^{-1} \boldsymbol{\Sigma})$. The factor $\sqrt{p}$ in (3.1) ensures that the overall magnitude of $\boldsymbol{\epsilon}_i$ stays bounded as $p \rightarrow \infty$, i.e., $\mathbb{E} \|\boldsymbol{\epsilon}_i\|^2 = O(1)$. When a random variable $\mathbf{X}$ is sub-Gaussian with parameter $\boldsymbol{\Sigma}$, note that $\boldsymbol{\Sigma}$ is not necessarily the true covariance matrix. It is a covariance proxy, a parameter that controls tail behavior via the moment generating function, as illustrated in the definition below.

Definition 1 (Sub-Gaussian random variable) A random variable $\mathbf{X}$ with $\mathbb{E}[\mathbf{X}] = 0$ is called sub-Gaussian with covariance proxy $\boldsymbol{\Sigma}$ if

$$\mathbb{E} \left[e^{\mathbf{t}^\top \mathbf{X}} \right] \leq \exp \left(\frac{\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}}{2} \right) \quad \text{for all } \mathbf{t} \in \mathbb{R}^p.$$

The centered normal distribution and any centered bounded distributions belong to the sub-Gaussian distribution family (Rigollet and Hütter, 2015). Using an appropriate bounded transformation, it would be possible to apply this method to any data coming from an unknown distribution F .

To establish our theoretical results, we consider the following controlled-growth-rate assumptions, as in Couillet and Benaych-Georges (2016) and Couillet et al. (2018).

Assumption 1 (*Growth Rate*) *As $n, p \rightarrow \infty$ the following conditions hold*

- *Data Scaling:* $\frac{p}{n} \rightarrow \bar{c}_0 > 0$ with $0 < \liminf_n \bar{c}_0 \leq \limsup_n \bar{c}_0 < \infty$
- *Mean Scaling:* $\|\boldsymbol{\mu}\| = O(1)$
- *Covariance scaling:* $\|\boldsymbol{\Sigma}\| = O(1)$ and $\text{tr}(\boldsymbol{\Sigma}) = O(p)$.

In the high-dimensional asymptotic regime where both the sample size (n) and dimension (p) diverge, we assume that their ratio stabilizes, i.e., $p/n \rightarrow \bar{c}_0 \in (0, \infty)$. The mean vector is required to remain bounded in Euclidean norm, $\|\boldsymbol{\mu}\| = O(1)$, preventing it from dominating the limiting behavior of quadratic forms. The covariance matrix $\boldsymbol{\Sigma}$ is similarly assumed to have bounded operator norm, ensuring the absence of exploding variance directions, while its total variance grows proportionally with the dimension p . Together, these conditions place the model firmly within the classical Marchenko-Pastur high-dimensional framework (Bai and Silverstein, 2010), guaranteeing stable spectral limits and enabling the application of random matrix theory to derive asymptotic results.

From (2.4) it is clear that the main diagonal elements of the kernel matrix will be all equal to $f(0)$, while the off-diagonal elements converge almost surely to $f(\tau^\circ)$, where τ° is defined below. In this article, our primary focus will be on the asymptotic behavior of the LS-SVR regression model under Assumptions 1 and 2, as well as the distribution given in (3.1).

Couillet and Benaych-Georges (2016) considered the random variables $\mathbf{X}_i$ to be normally distributed, each characterized by distinct mean vectors and covariance matrices. Next, we will demonstrate that a similar asymptotically linearized kernel matrix $\boldsymbol{\Omega}$ can be obtained for observations consisting of centered random variables belonging to the class of sub-Gaussian distributions, provided they have fixed mean vectors and covariance matrices. As such, our result in Theorem 3 is a significant relaxation of the normality assumption. To simplify notations throughout the paper, we introduce the following notations

$$\bar{\boldsymbol{\epsilon}} = \frac{1}{n} \sum_{i=1}^n \boldsymbol{\epsilon}_i, \quad \boldsymbol{\omega}_i = \boldsymbol{\epsilon}_i - \bar{\boldsymbol{\epsilon}}, \quad \psi_i = \|\boldsymbol{\omega}_i\|^2 - \mathbb{E} \|\boldsymbol{\omega}_i\|^2, \quad \tau^\circ = \frac{2}{p} \frac{n-1}{n} \text{tr}(\boldsymbol{\Sigma}),$$

$$\mathbf{P} = \mathbf{I}_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top \in \mathbb{R}^{n \times n}, \quad \mathbf{W} = [\boldsymbol{\omega}_1, \boldsymbol{\omega}_2, \dots, \boldsymbol{\omega}_n] \in \mathbb{R}^{p \times n}, \quad \boldsymbol{\psi} = [\psi_1, \psi_2, \dots, \psi_n]^\top \in \mathbb{R}^{n \times 1}.$$

Let the kernel function $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ satisfy the following assumption.

Assumption 2 (*Kernel function*) *The function f is a three-times differentiable function in a neighborhood of τ° .*

Theorem 1 provides uniform probability bounds for the quantities $\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j$ and ψ_i .

Theorem 1 Let $\mathbf{X}_i = \boldsymbol{\mu} + \sqrt{p}\boldsymbol{\epsilon}_i$, where $\boldsymbol{\epsilon}_i \stackrel{\text{ind}}{\sim} \text{subG}(p^{-1}\boldsymbol{\Sigma})$, then for $\delta \in (0, 1)$

$$\mathbb{P}\left(\max_{i \neq j} |\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j| \leq K_\star^2 \min \left\{ \sqrt{\frac{1}{c_1} \frac{\text{tr}(\boldsymbol{\Sigma}^2)}{p^2} \log \frac{2n^2}{\delta}}, \frac{1}{c_1} \frac{\|\boldsymbol{\Sigma}\|}{p} \log \frac{2n^2}{\delta} \right\}\right) \geq 1 - \delta$$

and

$$\mathbb{P}\left(\max_{1 \leq i \leq n} |\psi_i| \leq K_\star^2 \min \left\{ \sqrt{\frac{1}{c_2} \frac{\text{tr}(\boldsymbol{\Sigma}^2)}{p^2} \log \frac{2n}{\delta}}, \frac{1}{c_2} \frac{\|\boldsymbol{\Sigma}\|}{p} \log \frac{2n}{\delta} \right\}\right) \geq 1 - \delta,$$

where $c_1 > 0, c_2 > 0$ are constants and $K_\star$ (uniformly in n and p) is the sub-Gaussian constant.

Theorem 1 provides general uniform bounds for the quantities $\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j$ and ψ_i . However, one might be interested in more tractable uniform stochastic bounds under certain assumptions. Corollary 1 provides such uniform stochastic orders given our assumptions on the Frobenius and operator norms. The next corollary easily follows from Theorem 1, Borel-Cantelli and Assumption 1 i.e., $\|\boldsymbol{\Sigma}\| = O(1)$ and $\text{tr}(\boldsymbol{\Sigma}^2) = \sum_{i=1}^n \lambda_i^2 \leq \|\boldsymbol{\Sigma}\| \text{tr}(\boldsymbol{\Sigma})$, where λ_i denotes the i th eigenvalue of $\boldsymbol{\Sigma}$.

Corollary 1 Under Assumption 1, we have the following uniform bounds almost surely

$$\max_{i \neq j} |\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j| = O\left(\sqrt{\frac{\log n}{p}}\right) \quad \text{and} \quad \max_{1 \leq i \leq n} |\psi_i| = O\left(\sqrt{\frac{\log n}{p}}\right).$$

Depending on the assumptions, there are commonly three useful regimes. Then, the following bounds hold almost surely

- if $p \rightarrow \infty$ and $\log n = o(p)$ then

$$\max_{i \neq j} |\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j| = o(1) \quad \text{and} \quad \max_{1 \leq i \leq n} |\psi_i| = o(1),$$

- p is polynomial in n : $p = O(n^\rho)$ with $\rho > 0$, then

$$\max_{i \neq j} |\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j| = O\left(n^{-\rho/2} \sqrt{\log n}\right) \quad \text{and} \quad \max_{1 \leq i \leq n} |\psi_i| = O\left(n^{-\rho/2} \sqrt{\log n}\right),$$

- for $p = O(n \log n)$, then the bound is $O(n^{-1/2})$.

One of the main goals of this paper is to develop a random matrix approximation to the kernel matrix $\boldsymbol{\Omega}$ based on kernel functions of the form (2.4) via a Taylor expansion. A decomposition of the Euclidean norm gives (see Proof Theorem 3 in Appendix A)

$$\frac{1}{p} \|\mathbf{X}_i - \mathbf{X}_j\|^2 = \psi_i + \psi_j - 2\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j + \tau^\circ,$$

where ψ_i and ψ_j are random fluctuations of the squared norms of $\boldsymbol{\omega}_i$ and $\boldsymbol{\omega}_j$ around their expected values respectively, $2\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j$ captures the correlation between $\boldsymbol{\omega}_i$ and $\boldsymbol{\omega}_j$ and, $\tau^\circ = 2\mathbb{E}\|\boldsymbol{\omega}_j\|^2$ is twice the expected squared distance of the vector $\boldsymbol{\omega}_j$. From this expansion, it

is clear that the first three terms are stochastic while the last one is deterministic. The last term will be shown to be important in the Taylor expansion of the kernel matrix $\mathbf{\Omega}$. A quantity of particular interest in high-dimensional statistics is

$$\frac{1}{p} \|\mathbf{X}_i - \mathbf{X}_j\|^2 - \tau^\circ = \psi_i + \psi_j - 2\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j.$$

The next theorem illustrates a probabilistic bound on this quantity. According to Theorem 2 (an extension of Couillet and Benaych-Georges (2016) for sub-Gaussian random variables) the random pairwise distances are tightly concentrated around τ° in high dimensions and become asymptotically identical. In other words, all points are equally far apart in high-dimensional spaces, up to small fluctuations. This is a key phenomenon in random matrix theory and high-dimensional geometry.

Theorem 2 *Under Assumption 1 and let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \in \mathbb{R}^p$ be independent vectors. Further assume*

$$\mathbf{X}_i = \boldsymbol{\mu} + \sqrt{p}\boldsymbol{\epsilon}_i$$

for some $\boldsymbol{\mu} \in \mathbb{R}^p$, $\boldsymbol{\epsilon}_i \stackrel{\text{ind}}{\sim} \text{subG}(p^{-1}\boldsymbol{\Sigma})$ and $\mathbb{E}[\boldsymbol{\epsilon}_i] = 0$ such that $\mathbb{E}[\mathbf{X}_i] = \boldsymbol{\mu}$, then for any fixed $1 \leq i \neq j \leq n$ and $n, p \rightarrow \infty$

$$\frac{1}{p} \|\mathbf{X}_i - \mathbf{X}_j\|^2 \xrightarrow{\text{a.s.}} \tau^\circ \quad \text{and} \quad \max_{i \neq j} \frac{1}{p} \|\mathbf{X}_i - \mathbf{X}_j\|^2 \xrightarrow{\text{a.s.}} \tau^\circ.$$

Further, the following convergence rates hold almost surely

$$\frac{1}{p} \|\mathbf{X}_i - \mathbf{X}_j\|^2 = \tau^\circ + O\left(\sqrt{\frac{\log n}{p}}\right) \quad \text{and} \quad \max_{i \neq j} \frac{1}{p} \|\mathbf{X}_i - \mathbf{X}_j\|^2 = \tau^\circ + O\left(\sqrt{\frac{\log n}{p}}\right).$$

Remark 2 *It can be shown that $\tau^\circ = 2\mathbb{E}\|\boldsymbol{\omega}_j\|^2 = 2\frac{n-1}{n}\frac{1}{p}\text{tr}(\boldsymbol{\Sigma})$. For n large, τ° can be replaced in Theorem 2 by $\tau = 2\frac{\text{tr}(\boldsymbol{\Sigma})}{p}$ as $\tau^\circ \xrightarrow{\text{a.s.}} \tau$. Although the order of the pointwise and uniform rates is the same, the constants are quite different. In fact, the constant depends on $\bar{c}_0$ and the rate degrades as $\bar{c}_0 \rightarrow \infty$, i.e. $p \gg n$. This indicates that high-dimensional data relative to sample size inflates fluctuations, consistent with the Marchenko-Pastur law.*

Theorem 3 provides an approximation to random kernel matrices of the form (2.4) for sub-Gaussian random variables via a Taylor series around τ° . This theorem relaxes the conditions of El Karoui (2010) and Couillet and Benaych-Georges (2016) to the sub-Gaussian case.

Theorem 3 *Under Assumption 1 and 2 and let the kernel K be defined as in (2.4). Then as $n \rightarrow \infty$, the full random kernel matrix expansion is*

$$\begin{aligned} \mathbf{\Omega} &= -2f'(\tau^\circ)(\mathbf{P}\mathbf{W}^\top \mathbf{W}\mathbf{P} + \mathbf{A}) + \beta \mathbf{I}_n + O(n^{-1/2}) \text{ a.s.} \\ &= \tilde{\mathbf{\Omega}} + O(n^{-1/2}) \text{ a.s.} \end{aligned}$$

where

$$\begin{aligned} \beta &= f(0) - f(\tau^\circ) + \tau^\circ f'(\tau^\circ) \\ \mathbf{A} &= \mathbf{A}_n + \mathbf{A}_{\sqrt{n}} + \mathbf{A}_1 \end{aligned}$$

and

$$\begin{aligned}\mathbf{A}_n &= -\frac{f(\tau^\circ)}{2f'(\tau^\circ)}\mathbf{1}_n\mathbf{1}_n^\top \\ \mathbf{A}_{\sqrt{n}} &= -\frac{1}{2}(\boldsymbol{\psi}\mathbf{1}_n^\top + \mathbf{1}_n\boldsymbol{\psi}^\top) \\ \mathbf{A}_1 &= -\frac{f''(\tau^\circ)}{4f'(\tau^\circ)}(\boldsymbol{\psi}^2\mathbf{1}_n^\top + \mathbf{1}_n\boldsymbol{\psi}^2^\top + \frac{4}{p^2}\|\boldsymbol{\Sigma}\|_F^2\mathbf{1}_n\mathbf{1}_n^\top + 2\boldsymbol{\psi}\boldsymbol{\psi}^\top)\end{aligned}$$

An immediate consequence of Theorem 3 is that the random kernel vector approximation at an arbitrary point $\mathbf{x}$, as defined in (2.5), is given in the next corollary.

Corollary 2 *The approximate kernel vector in an arbitrary point $\mathbf{x}$ is*

$$\begin{aligned}k(\mathbf{x}) &= f(\tau^\circ)\mathbf{1}_n + f'(\tau^\circ)(\boldsymbol{\psi}_\mathbf{x}\mathbf{1}_n + \boldsymbol{\psi} - 2\mathbf{P}\mathbf{W}^\top\boldsymbol{\epsilon}_\mathbf{x}) + \frac{f''(\tau^\circ)}{2}(\boldsymbol{\psi}_\mathbf{x}^2\mathbf{1}_n + \boldsymbol{\psi}^2 + 2\boldsymbol{\psi}_\mathbf{x}\boldsymbol{\psi} + \frac{4}{p^2}\|\boldsymbol{\Sigma}\|_F^2\mathbf{1}_n) \\ &\quad + (f(0) - f(\tau^\circ) + \tau^\circ f'(\tau^\circ))\mathbf{I}\{\mathbf{X} = \mathbf{x}\} + O(n^{-1/2}) \text{ a.s.},\end{aligned}$$

where $\mathbf{I}\{\mathbf{X} = \mathbf{x}\}$ is an indicator function yielding 1 when $\mathbf{X} = \mathbf{x}$, 0 otherwise and $\boldsymbol{\psi}_\mathbf{x} = \|\boldsymbol{\epsilon}_\mathbf{x}\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_\mathbf{x}\|^2$ with $\boldsymbol{\epsilon}_\mathbf{x} = (\mathbf{x} - \boldsymbol{\mu})/\sqrt{p}$.

Note that Corollary 2 involves the term $\|\boldsymbol{\Sigma}\|_F^2$, which requires estimating the covariance matrix. In high dimensions, where the number of variables is comparable to or exceeds the sample size, the sample covariance becomes singular or ill-conditioned, and its eigenvalues are distorted by noise, destabilizing analyses such as PCA and discriminant analysis. Modern approaches address this via regularization techniques—such as shrinkage, sparsity, banding, and factor models—that impose structure and ensure stable, consistent estimation (Zhang et al., 2023). For recent developments in high-dimensional covariance estimation, we refer the reader to Appendix E.

Theorem 4 illustrates the quality of the approximation for the kernel matrix $\boldsymbol{\Omega}$ via $\tilde{\boldsymbol{\Omega}}$. In other words, the difference between the matrices $\boldsymbol{\Omega}$ and $\tilde{\boldsymbol{\Omega}}$ measured in operator norm converges to zero as n and p grow according to Assumption 1.

Theorem 4 *Under the assumptions of Theorem 2 and Theorem 3, then as $n, p \rightarrow \infty$*

$$\|\boldsymbol{\Omega} - \tilde{\boldsymbol{\Omega}}\| \xrightarrow{\text{a.s.}} 0,$$

in operator norm, where $\tilde{\boldsymbol{\Omega}} = -2f'(\tau^\circ)(\mathbf{P}\mathbf{W}^\top\mathbf{W}\mathbf{P} + \mathbf{A}) + \beta\mathbf{I}_n$.

According to the Hoffman-Wielandt inequality (Horn and Johnson, 2013, Theorem 6.3.5), it immediately follows from Theorem 4 that

$$\max_i |\lambda_i(\boldsymbol{\Omega}) - \lambda_i(\tilde{\boldsymbol{\Omega}})| \xrightarrow{\text{a.s.}} 0.$$

where $\lambda_i(\boldsymbol{\Omega})$ denotes the i th eigenvalue of the matrix $\boldsymbol{\Omega}$. This implies that the determination of isolated eigenvalues in the spectrum of $\boldsymbol{\Omega}$ can be studied from the equivalent problem for $\tilde{\boldsymbol{\Omega}}$. This is an important topic in kernel spectral clustering (see e.g. Couillet and Benaych-Georges (2016)), but will not be further explored in this paper.

Figure 1(a) illustrates the kernel density estimate for each matrix entry for the exact and approximate (based on Theorem 3) RBF kernel matrix based on data generated from a multivariate normal distribution with $n = 256$, $p = 512$, mean vector 5, covariance matrix $\Sigma_{ij} = 0.4^{|i-j|}(1 + 5/\sqrt{p})$ for $i, j = 1, \dots, p$, and $1/h^2 = 1$. Figure 1(b) shows the eigenvalue spectrum (first 30 eigenvalues) for the exact and approximate RBF kernel matrix. Figure 2(a) and (b) illustrate the same principle, but based on data generated from a uniform $(0, 1)$ distribution with $n = 256$, $p = 512$, and $1/h^2 = 10$. From this example, it follows that, for the RBF kernel, the approximate kernel matrix derived from Theorem 3 closely matches the exact kernel matrix for both Gaussian and non-Gaussian (uniformly distributed) data. For visual clarity, we did not include the largest eigenvalue in either bar chart, but it is noted below the figure.

Since non-Gaussian random variables can often be transformed into bounded variables—for instance, through min-max (0–1) normalization—the proposed approximation remains applicable to a wide class of non-Gaussian data after appropriate pre-processing.

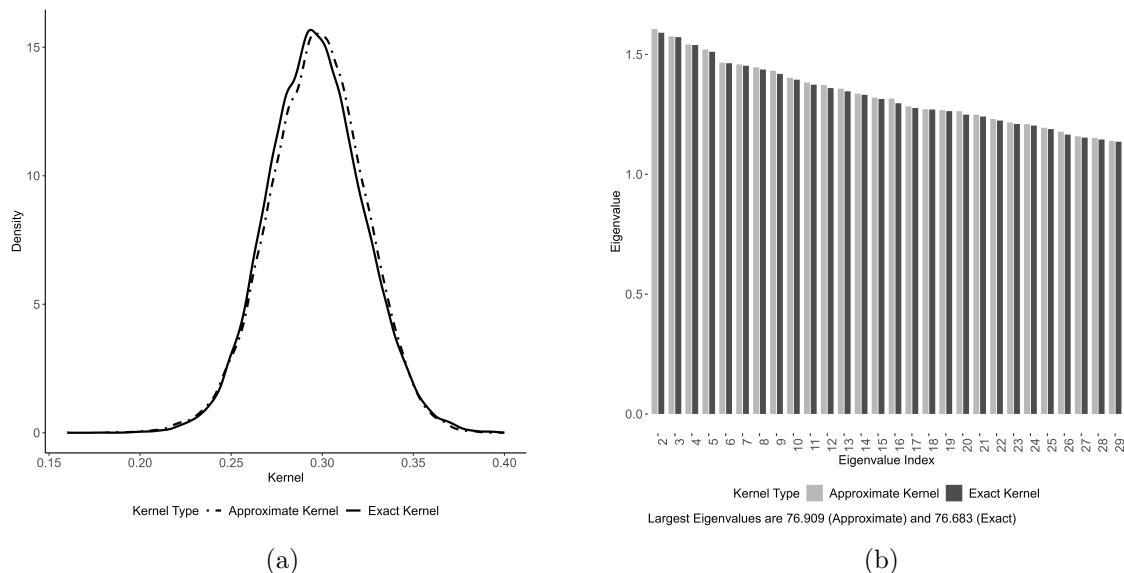


Figure 1: Data generated from multivariate normal distribution with mean = 5, $\Sigma_{ij} = 0.4^{|i-j|}(1 + 5/\sqrt{p})$ for $i, j = 1, \dots, p$ and $1/h^2 = 1$. (a) Density of the exact kernel matrix elements and their approximations for $n = 256$, and $p = 512$. (b) Eigenvalue spectrum (first 30 eigenvalues) of the exact kernel matrix elements and its approximation based on Theorem 3. For clarity, the largest eigenvalue is not shown; it is noted below the bar chart.

It is important to note, however, that the quality of the kernel approximation depends critically on the bandwidth parameter h . In Theorem 3, the bandwidth is assumed to satisfy $h = O(1)$. Although beyond the scope of this paper, considering the dependence of the bandwidth (and regularization parameter) on n and p is an interesting topic for further research.

Using Theorem 3 and a Taylor expansion around τ° , an approximation of $\mathbf{S}^{-1}$ can now be obtained. Additionally, using the approximation for $\mathbf{S}^{-1}$, it is possible to find approximations for $\hat{\alpha}$ and $\hat{b}$, and finally obtain an asymptotic approximation for $\hat{g}(\mathbf{x})$.

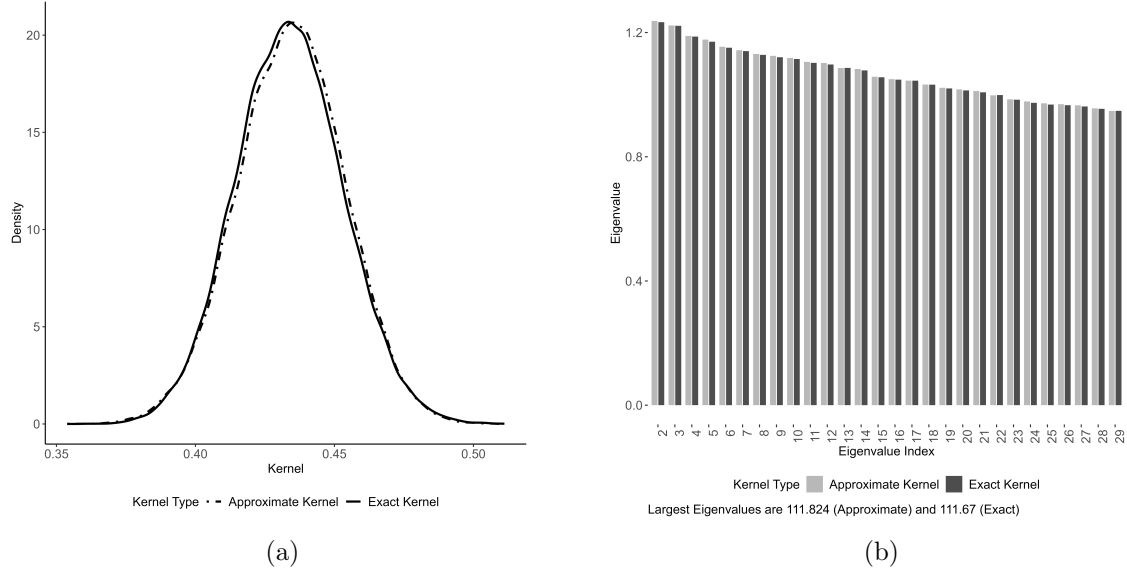


Figure 2: Data generated from the uniform distribution $(0, 1)$ and $1/h^2 = 10$. (a) Kernel density estimate of the exact kernel matrix elements and their approximations for $n = 256$, and $p = 512$. (b) Eigenvalue spectrum (first 30 eigenvalues) of the exact kernel matrix and its approximation based on Theorem 3. For clarity, the largest eigenvalue is not shown; it is noted below the bar chart.

Theorem 5 illustrates the large-dimensional approximation of the LS-SVR regression model under the sub-Gaussian assumption. Similarly, but under more stringent assumptions, Liao and Couillet (2019) showed that the LS-SVM classification model admits a large-dimensional approximation. To stress the dependence on the bandwidth h under which the Neumann series expansions for matrices converge absolutely in operator norm almost surely, we have used $f(\frac{\tau^\circ}{2h^2})$ instead of $f(\tau^\circ)$ in Theorem 5.

Theorem 5 (Asymptotic Approximation) *Let Assumptions 1 and 2 hold. Further, let $\mathbf{X} = \boldsymbol{\mu} + \sqrt{p}\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon} \sim \text{subG}(p^{-1}\boldsymbol{\Sigma})$, $f(\frac{\tau^\circ}{2h^2}) < 1/\gamma$, and $\hat{g}(\mathbf{x})$ be defined by (2.3). Then, as $n, p \rightarrow \infty$,*

$$\tilde{g}(\mathbf{x}) \xrightarrow{a.s.} \hat{g}(\mathbf{x}),$$

with

$$\begin{aligned} \tilde{g}(\mathbf{x}) &= \bar{Y} - \frac{2\gamma f'(\tau^\circ)}{n} \mathbf{Y}^\top \mathbf{W}^\top \boldsymbol{\epsilon}_\mathbf{x} + \frac{\gamma f''(\tau^\circ)}{n} \mathbf{Y}^\top \boldsymbol{\psi} \boldsymbol{\psi}_\mathbf{x} \\ &= \sum_{i=1}^n \left\{ \frac{1}{n} - \frac{2\gamma f'(\tau^\circ)}{n} \omega_i^\top \boldsymbol{\epsilon}_\mathbf{x} + \frac{\gamma f''(\tau^\circ)}{n} \psi_i \psi_x \right\} Y_i, \end{aligned} \quad (3.2)$$

where $\boldsymbol{\psi}_\mathbf{x} = \|\boldsymbol{\epsilon}_\mathbf{x}\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_\mathbf{x}\|^2$ and $\boldsymbol{\epsilon}_\mathbf{x} = (\mathbf{x} - \boldsymbol{\mu})/\sqrt{p}$.

Remark 3 *Theorem 5 relies on technical Lemma 3 given in Appendix A, which states the conditions under which the Neumann series expansions for matrices converge absolutely*

in operator norm almost surely. Figure 3 illustrates matrix series expansions in high-dimensional random matrices: the blue arrow shows the Neumann series $(\mathbf{I} + \frac{\gamma}{n}\mathbf{\Omega})^{-1} = \sum (-1)^k (\frac{\gamma}{n}\mathbf{\Omega})^k$ for bulk eigenvalues $|\lambda_i| < 1$, capturing small-norm fluctuations, while the red arrow shows the inverse expansion $(\mathbf{I} - \frac{\gamma}{n}\mathbf{\Omega})^{-1} = -\sum (\frac{\gamma}{n}\mathbf{\Omega})^{-k}$ for large eigenvalues $\lambda_i > 1$, corresponding to spikes or low-rank signals. This separation illustrates the random matrix theory perspective of bulk versus spike components in kernel or covariance matrices (Bai and Silverstein (2010) and Benaych-Georges and Nadakuditi (2011)). Under the condition $p = O(n)$, we show, in technical Lemma 3 in Appendix A, that $\|\mathbf{\Omega}\| = nf(\frac{\sigma^2}{2h^2}) + O(1) < \frac{n}{\gamma}$ a.s. However, the $O(1)$ cannot be made $o(1)$ unless $p = o(n)$.

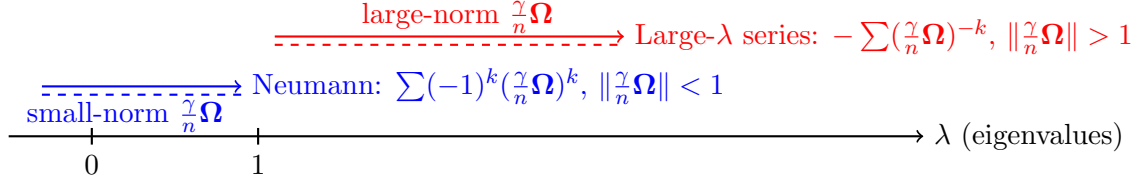


Figure 3: Eigenvalue-based illustration of matrix series expansions: the blue arrow shows the Neumann series $(\mathbf{I} + \frac{\gamma}{n}\mathbf{\Omega})^{-1} = \sum (-1)^k (\frac{\gamma}{n}\mathbf{\Omega})^k$ for small-norm matrices ($\|\frac{\gamma}{n}\mathbf{\Omega}\| < 1$), while the red arrow shows the large-eigenvalue expansion $(\mathbf{I} - \frac{\gamma}{n}\mathbf{\Omega})^{-1} = -\sum (\frac{\gamma}{n}\mathbf{\Omega})^{-k}$ for matrices with $\|\frac{\gamma}{n}\mathbf{\Omega}\| > 1$. Bulk eigenvalues correspond to small-norm components, and spike eigenvalues correspond to large-norm components.

In Theorem 5, the term $\bar{Y}$ represents the mean value of the responses $\mathbf{Y}$ and is $O_p(1)$. The other two terms describe the noise involved in the approximation. The second term of (3.2) encompasses the sum of the noise from new values as well as the covariance between the response $\mathbf{Y}$ and the noise from the input data $\epsilon_{\mathbf{x}}$. This formulation implies that a stronger correlation between the response variable and the input variable significantly affects the estimation of $\tilde{g}(\mathbf{X})$. Higher correlation amplifies the contribution of the input's noise to the prediction, which can potentially move $\tilde{g}(\mathbf{X})$ further away from $\bar{Y}$, especially when $\epsilon_{\mathbf{x}}$ deviates more from zero. Therefore, this relationship underscores that the accuracy of the estimation relies on both the strength of correlation and the magnitude of noise in the input, which is essential for understanding the precision and accuracy of $\tilde{g}(\mathbf{X})$.

The third term in (3.2) involves the product of the centered norm of the new observation and the covariance between the response variable and the squared norms of the input observations. This covariance remains constant across all new observations, indicating that the centered norm of each new observation solely determines its impact. The squared norms of the input observations represent the variance within the p -dimensional input space. This term reflects how variations in the magnitude of new observations, centered around their mean, interact with the covariance structure between the response and the overall variability present within the input space. Consequently, it elucidates how changes in the norms of new observations, relative to their mean values, affect the estimation process, encapsulating the intrinsic variability inherent in the input data dimensions.

In the context of LS-SVR, γ is a regularization parameter that balances the trade-off between minimizing the training error and reducing the model complexity. The data dimensionality and the selection of γ are interdependent and influence the model's performance.

However, in (3.2), $\tilde{g}(\mathbf{X})$ is proportional to the hyperparameter γ , indicating a linear relationship between $\tilde{g}(\mathbf{X})$ and the noise term. For any large value of γ , the estimate $\tilde{g}(\mathbf{X})$ will deviate from the mean $\bar{Y}$, while a small value of γ will cause the estimate to converge towards $\bar{Y}$.

For the RBF kernel, we have $f(\tau^\circ) > 0$, $f'(\tau^\circ) < 0$, and $f''(\tau^\circ) > 0$. Since τ° is fixed based on the given data, a very large value of h results in $f'(\tau^\circ) = -\frac{1}{2h^2} \exp\{-\tau^\circ/(2h^2)\} < 0$ for RBF kernel, resulting in a small magnitude. This, coupled with $f''(\tau^\circ) > 0$, implies there will be minimal deviation from the mean $\bar{Y}$ when estimating the function $\tilde{g}(\mathbf{X})$. Therefore, both the regularization parameter γ , the first and second order derivative of the kernel function at τ° play critical roles in determining how closely $\tilde{g}(\mathbf{X})$ aligns with the sample mean $\bar{Y}$, where larger values of γ and smaller magnitudes of $f'(\tau^\circ)$ increase the deviation from $\bar{Y}$.

To use Theorems 3 and 5 in practice, we need an estimator for τ° . Under the sub-Gaussian assumption, τ° can be easily estimated in the large n and p regime, as Theorem 6 suggests. Since the proof is short, we provide it in the main text rather than in the Appendix.

Theorem 6 *Under Assumption 1 as $n \rightarrow \infty$,*

$$\hat{\tau}^\circ = \frac{2}{n} \sum_{i=1}^n \frac{\|\mathbf{X}_i - \bar{\mathbf{X}}\|^2}{p} \xrightarrow{a.s.} \tau = \frac{2}{p} \text{tr}(\mathbf{\Sigma})$$

where $\bar{\mathbf{X}} = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i$.

Proof. By expanding the norm and using $\mathbf{X}_i = \boldsymbol{\mu} + \sqrt{p}\boldsymbol{\epsilon}_i$

$$\frac{2}{n} \sum_{i=1}^n \frac{\|\mathbf{X}_i - \bar{\mathbf{X}}\|^2}{p} = \frac{2}{n} \sum_{i=1}^n \frac{\|\boldsymbol{\mu} + \sqrt{p}\boldsymbol{\epsilon}_i - \boldsymbol{\mu} - \sqrt{p}\bar{\boldsymbol{\epsilon}}\|^2}{p} = \frac{2}{n} \sum_{i=1}^n \|\boldsymbol{\epsilon}_i\|^2 - 2\|\bar{\boldsymbol{\epsilon}}\|^2.$$

Since $\boldsymbol{\epsilon}_i \sim \text{subG}(p^{-1}\mathbf{\Sigma})$, then by the strong law of large numbers

$$\frac{1}{n} \sum_{i=1}^n \|\boldsymbol{\epsilon}_i\|^2 \rightarrow \mathbb{E}\|\boldsymbol{\epsilon}_1\|^2 \text{ a.s.}$$

Since $\|\boldsymbol{\epsilon}_i\|^2$ is a quadratic form it immediately follows that $\mathbb{E}\|\boldsymbol{\epsilon}_1\|^2 = p^{-1}\text{tr}(\mathbf{\Sigma})$. For the second term, note that $\bar{\boldsymbol{\epsilon}} \sim \text{subG}(n^{-1}p^{-1}\mathbf{\Sigma})$ and therefore by the Hanson-Wright inequality (see Lemma 1 in Appendix A and Hanson and Wright (1971)) it follows that $\|\bar{\boldsymbol{\epsilon}}\|^2 \rightarrow \mathbb{E}\|\bar{\boldsymbol{\epsilon}}\|^2$ a.s. The result follows by combining the two terms, $\text{tr}(\mathbf{\Sigma}) = O(p)$ (by Assumption 1) and letting $n \rightarrow \infty$, then $\mathbb{E}\|\bar{\boldsymbol{\epsilon}}\|^2 = \frac{\text{tr}(\mathbf{\Sigma})}{np} \rightarrow 0$ a.s. $\blacksquare$

4. Asymptotic Normality of the Large Scale Approximation

To understand the large-sample behavior of the approximate estimator, we derive its asymptotic distribution under some mild regularity conditions. This analysis provides theoretical guarantees for the estimator's consistency and supports statistical inference. In particular, we establish the limiting distribution as the sample size tends to infinity, highlighting the

influence of the model parameters and the structure of the input data. The results presented here form the foundation for evaluating the estimator's variability in high-dimensional settings.

Theorem 7 (Asymptotic Normality) *Let $\xi_1, \dots, \xi_n$, with $\xi_k = l_k(\mathbf{x})Y_k$, be a sequence of independent random variables with finite second moments. Let $\text{Var}[\xi_k|\mathbb{X}] < \infty$ and $\tilde{g}(\mathbf{x}) = \xi_1 + \dots + \xi_n$, then*

$$\frac{\tilde{g}(\mathbf{x}) - \mathbb{E}(\tilde{g}(\mathbf{x}) | \mathbb{X})}{\sqrt{\text{Var}(\tilde{g}(\mathbf{x}) | \mathbb{X})}} \xrightarrow{d} N(0, 1).$$

The proof of Theorem 7 is given in Appendix H.

Remark 4 *It can be shown that $\hat{b}$ and $\hat{\alpha}$ obtained in (G.6) and (G.7), respectively, follow a normal distribution and*

$$\hat{b} = \bar{Y} - \frac{\gamma f'(\tau^\circ)}{n} \boldsymbol{\psi}^\top \mathbf{P} \mathbf{Y} - \frac{\gamma f''(\tau^\circ)}{2n} \boldsymbol{\psi}^2 \mathbf{P} \mathbf{Y} + O(n^{-3/2}) \text{ a.s.}$$

and

$$\hat{\alpha} = \frac{\gamma}{n} \mathbf{P} \mathbf{Y} + \gamma \mathbf{P} \left(\mathbf{Q} - \frac{\beta}{n^2} \mathbf{I}_n \right) \mathbf{P} \mathbf{Y} - \frac{\gamma^3 (f'(\tau^\circ))^2}{n^2 (1 + \gamma f(\tau^\circ))} \boldsymbol{\psi} \boldsymbol{\psi}^\top \mathbf{P} \mathbf{Y} + O(n^{-5/2}) \text{ a.s.}$$

where

$$\mathbf{Q} = \frac{2f'(\tau^\circ)}{n^2} \left(\mathbf{A}_1 + \mathbf{P} \mathbf{W}^\top \mathbf{W} \mathbf{P} + \frac{2f'(\tau^\circ)}{n} \mathbf{A}_{\sqrt{n}} \mathbf{L} \mathbf{A}_{\sqrt{n}} \right).$$

5. Simulation

Consider the following two models

$$\mathbf{Y} = \mathbf{X} \boldsymbol{\beta} + e \quad (5.1)$$

$$\mathbf{Y} = \sin(\mathbf{X}) \boldsymbol{\beta} + e \quad (5.2)$$

where $\mathbf{X} \sim \text{MVN}(\boldsymbol{\mu}, \boldsymbol{\Omega})$ with $\boldsymbol{\mu} = \mathbf{0}$, $\boldsymbol{\beta} = [\beta_1, \dots, \beta_p]^\top$ is a vector of regression parameters where each $\beta_j \sim N(0, 1)$, $j = 1, \dots, p$, and $e \sim N(0, 1)$. Various parameter combinations and different covariance matrices are given in Table 1. Figure 4 shows the kernel density

$\boldsymbol{\Omega}$	n	p	$1/h^2$	γ
$\mathbf{I}_p$	20, 50,	20, 50,	0.001,	0.001,
$\text{AR}_p(1)$	100, 500	100, 500	1	1

Table 1: Various parameters considered for the simulation.

estimates of the LS-SVM estimates $\hat{g}(\mathbf{x})$ and their asymptotic approximations $\tilde{g}(\mathbf{x})$ for 100 new observations under (5.1) with $\text{AR}_p(1)$ covariance structure. The estimates are computed for varying n and p , with a fixed bandwidth and regularization parameter and illustrate the approximation accuracy. Larger n and p reduce the deviation between $\hat{g}(\mathbf{x})$ and $\tilde{g}(\mathbf{x})$. The absolute difference between $\hat{g}(\mathbf{x})$ and $\tilde{g}(\mathbf{x})$ decreases significantly as n and p increase.

Overall, these simulations illustrate the impact of sample size and dimensionality on the approximation accuracy, with larger n and p generally leading to better approximations. For brevity, we did not include more simulations in this section. However, for the interested reader, more simulations for both models with varying covariance structure can be found in Appendix I.

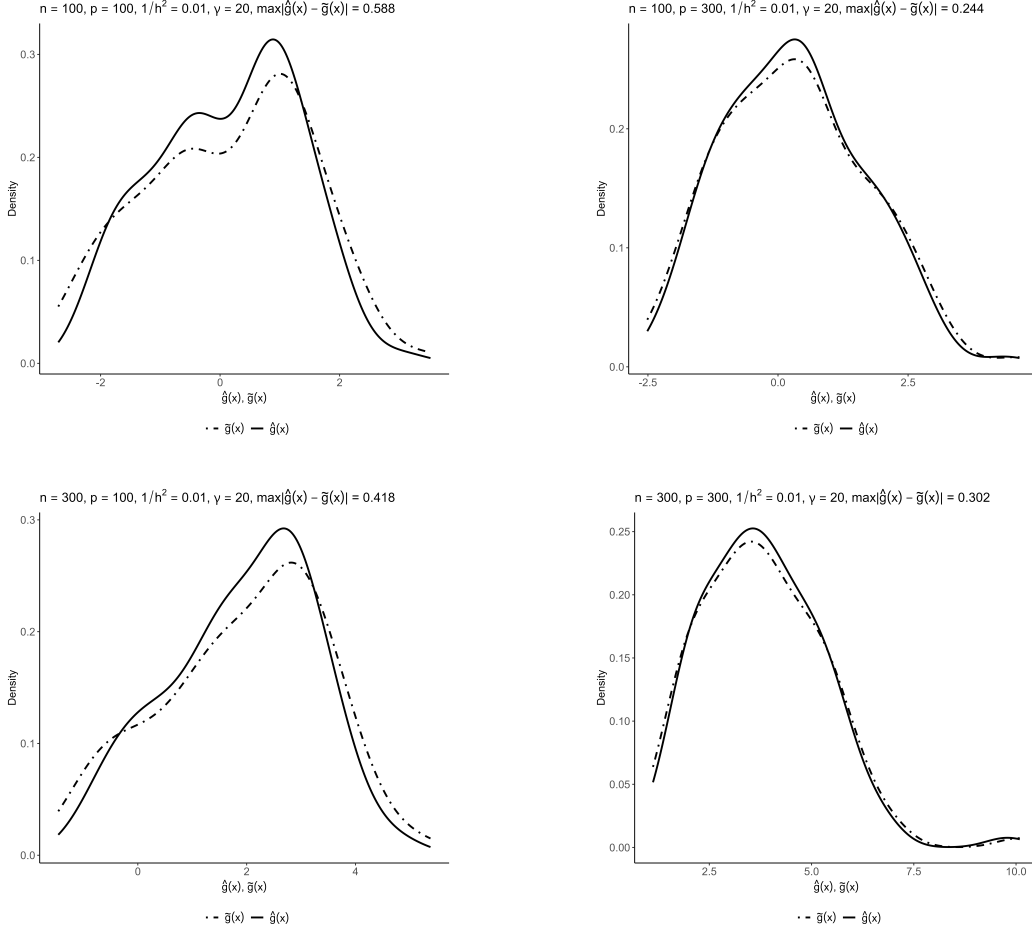


Figure 4: Kernel density estimate for $\hat{g}(\mathbf{x})$ (full) and $\tilde{g}(\mathbf{x})$ (dash dot) for 100 new observations for varying n and p combinations under (5.1) with $\text{AR}_p(1)$ covariance structure, $1/h^2 = 0.01$, and $\gamma = 20$.

6. Real Dataset Application

This study utilized the clinical pathology data from the Open TG-GATEs database (Igarashi et al., 2015), a publicly accessible resource dedicated to toxicological research. Open TG-GATEs encompasses toxicogenomic profiles alongside traditional toxicological data, sourced from both *in vivo* experiments (rat liver and kidney) and *in vitro* experiments (rat and human primary hepatocytes). These datasets represent exposure to 170 distinct compounds across a variety of dosages and time points. We focused on RBC($\times 10_4/\text{ul}$) clinical pathology

measurements, generated following standardized animal study protocols. Each experiment was conducted at three dose levels (low, medium, and high) and over four distinct treatment durations (3, 7, 14, and 28 days). The dataset comprises clinical pathology data from 8,078 rats treated with 138 compounds under 1,649 unique treatment conditions, each defined by the compound, dose, and duration. Chemical structure information for the 138 compounds was also incorporated as input for model development. The compound data, including PubChem CIDs, Structure-Data Files (SDF), and canonical SMILES strings, were retrieved using the PubChemPy Python package (Kim et al., 2021; Swain, 2014). Subsequently, molecular descriptors were calculated using the Mordred package (Moriwaki et al., 2018), generating a comprehensive set of 1,826 three-dimensional (3D) molecular descriptors for each compound.

To evaluate the performance of the approximation, we compared the approximate LS-SVR to the full LS-SVR under three bandwidth settings ($h^2 = 0.1, 1, 10$) with $\gamma = 1$. Overall, the approximate estimator closely matches the exact solution, particularly for moderate and large bandwidth values.

For a larger bandwidth ($h^2 = 10$), the approximation exhibits near-perfect agreement with the full LS-SVR, as evidenced by high correlations in both the training and test sets (0.95 and 0.90, respectively). Similarly strong performance was observed for $h^2 = 1$, although the correlations were slightly lower.

In contrast, for the small bandwidth ($h^2 = 0.1$), the approximation deteriorated. The model’s increased sensitivity led to greater variability between the approximation and the full LS-SVR, resulting in lower correlations (0.8645 for the training data and 0.7585 for the test data). This pattern underscores the critical role of the bandwidth parameter, which depends on both n and p . When the bandwidth is small, the approximation error becomes more pronounced. Conversely, larger bandwidth values induce greater smoothing, stabilize the estimation process, and allow the approximation to more closely track the full LS-SVR solution.

Figure 5 displays scatterplots of $\hat{g}(\mathbf{x})$ versus $\tilde{g}(\mathbf{x})$ across the three bandwidth settings for training data. Table 2 reports the corresponding Pearson’s correlation coefficients for the training and test data.

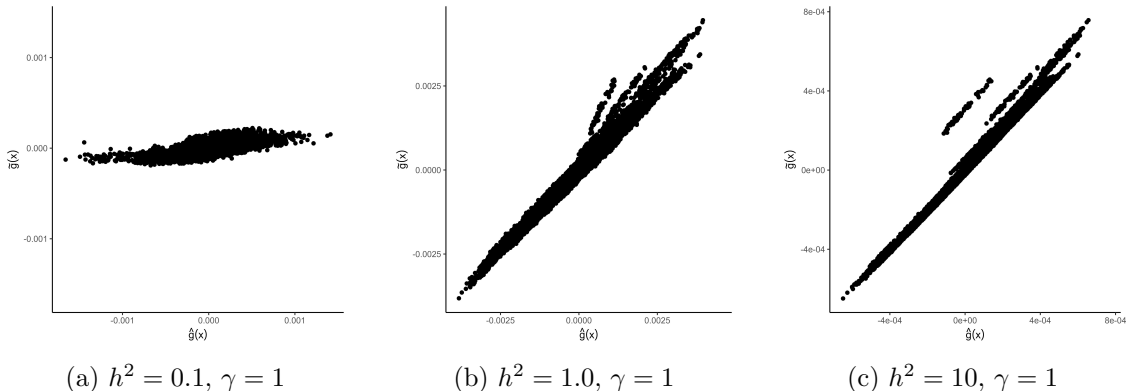


Figure 5: Scatter plot of $\hat{g}(\mathbf{x})$ and $\tilde{g}(\mathbf{x})$ for different h^2 .

h^2	γ	train correlation	test correlation
0.1	1	0.86	0.76
1	1	0.90	0.85
10	1	0.95	0.90

Table 2: Pearson’s correlation coefficients between $\hat{g}(\mathbf{x})$ and $\tilde{g}(\mathbf{x})$ for the training and test data.

7. Conclusion

This study investigates the asymptotic behavior of Least Squares Support Vector Regression (LS-SVR) in high-dimensional settings where both the number of observations (n) and features (p) grow to infinity while their ratio remains constant ($n/p \rightarrow \bar{c}_0$). Using kernel approximations combined with advanced asymptotic techniques, we derived explicit expressions for the LS-SVR function in this regime.

Our analysis shows that, under high-dimensional scaling, the regression coefficients and the resulting estimated regression function exhibit asymptotic normality. The results further reveal that the choice of kernel and its associated parameters, particularly the bandwidth, plays a critical role in determining model performance.

In summary, this work advances the theoretical understanding of LS-SVR in high-dimensional regimes and provides practical insight into its behavior, strengths, and limitations in large-scale applications.

Appendix A.

A. Technical lemmas

Before presenting the proof of Theorems 1 to 7, we will introduce two technical lemmas that are necessary for our theoretical results. Lemma 1 (Rudelson and Vershynin, 2013) is a Hanson-Wright concentration inequality (Hanson and Wright, 1971) for random vectors with independent sub-Gaussian coordinates. The main idea behind this inequality is quantification of the tension between collective fluctuations (controlled by $\|\mathbf{A}\Sigma_X\|_F^2$) and directional extremes (controlled by $\|\mathbf{A}\Sigma_X\|$), with the minimum selecting whichever source of randomness is the harder to control at the given deviation level t . Lemma 2 determines the stochastic order of two key quantities while Lemma 3 establishes the condition under which the Neumann series is valid (used to approximate the inverse of the sum of two matrices).

Lemma 1 (Hanson-Wright inequality) *Let $\mathbf{X} \in \mathbb{R}^p$ be a mean-zero sub-Gaussian vector with sub-Gaussian constant $K_\star = \max_{1 \leq j \leq p} \sup_{q \geq 1} q^{-1/2} (\mathbb{E}|\mathbf{X}_{1j}|^q)^{1/q}$ and covariance Σ_X , then for any fixed symmetric matrix $\mathbf{A}$,*

$$\mathbb{P}(|\mathbf{X}^\top \mathbf{A} \mathbf{X} - \mathbb{E}[\mathbf{X}^\top \mathbf{A} \mathbf{X}]| > t) \leq 2 \exp \left\{ -c \min \left(\frac{t^2}{K_\star^4 \|\mathbf{A}\Sigma_X\|_F^2}, \frac{t}{K_\star^2 \|\mathbf{A}\Sigma_X\|} \right) \right\},$$

where $c > 0$, $\|\cdot\|_F$ and $\|\cdot\|$ denote the Frobenius and operator norm respectively.

Lemma 2 *Let $\mathbf{X}_i = \boldsymbol{\mu} + \sqrt{p}\boldsymbol{\epsilon}_i$, where $\boldsymbol{\epsilon}_i \stackrel{\text{ind}}{\sim} \text{subG}(p^{-1}\Sigma)$ and define $\boldsymbol{\omega}_i = \boldsymbol{\epsilon}_i - \bar{\boldsymbol{\epsilon}}$, then under Assumption 1*

$$\psi_i = \|\boldsymbol{\omega}_i\|^2 - \mathbb{E}\|\boldsymbol{\omega}_i\|^2 = O_p(n^{-1/2}) \quad \text{and} \quad \boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j = O_p(n^{-1/2}).$$

Proof of Lemma 2. By expanding $\|\boldsymbol{\omega}_i\|^2$, we have

$$\|\boldsymbol{\omega}_i\|^2 = \|\boldsymbol{\epsilon}_i\|^2 - 2\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}} + \|\bar{\boldsymbol{\epsilon}}\|^2$$

and hence ψ_i can be written as

$$\psi_i = [\|\boldsymbol{\epsilon}_i\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_i\|^2] - 2[\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}} - \mathbb{E}(\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}})] + [\|\bar{\boldsymbol{\epsilon}}\|^2 - \mathbb{E}\|\bar{\boldsymbol{\epsilon}}\|^2]. \quad (\text{A.1})$$

In what follows, we will bound each term. The first term in (A.1), $\|\boldsymbol{\epsilon}_i\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_i\|^2$, can be written as $\boldsymbol{\epsilon}_i^\top \mathbf{I}_p \boldsymbol{\epsilon}_i - \mathbb{E}[\boldsymbol{\epsilon}_i^\top \mathbf{I}_p \boldsymbol{\epsilon}_i]$. This is a quadratic form and can be immediately used in Lemma 1, since $\boldsymbol{\epsilon}_i$ is sub-Gaussian with $\mathbf{A} = \mathbf{I}_p$ and $\Sigma_X = \Sigma/p$. Next, we bound the Frobenius and operator norms. Under Assumption 1, we have $\text{tr}(\Sigma) = O(p)$ and $\|\Sigma\| = O(1)$

$$\begin{aligned} \|\mathbf{A}\Sigma_X\|_F^2 &= \|\mathbf{I}_p \Sigma/p\|_F^2 = \frac{1}{p^2} \text{tr}(\Sigma^2) = O(1/p) \\ \|\mathbf{A}\Sigma_X\| &= \|\mathbf{I}_p \Sigma/p\| = \frac{1}{p} \|\Sigma\| = O(1/p), \end{aligned}$$

since $\|\Sigma\|_F^2 = \text{tr}(\Sigma\Sigma^\top) = \text{tr}(\Sigma^2) = \sum_{i=1}^n \lambda_i^2 \leq \max_i \lambda_i \sum_{i=1}^n \lambda_i = \|\Sigma\| \text{tr}(\Sigma)$. The Hanson-Wright inequality then becomes

$$\mathbb{P}(|\|\epsilon_i\|^2 - \mathbb{E}\|\epsilon_i\|^2| > t) \leq 2 \exp \left\{ -c \min(C_1 t^2 p, C_2 t p) \right\}$$

where c, C_1 and C_2 are positive constant and C_1 and C_2 depend on $K_\star$. Taking $t = Mn^{-1/2}$ with $M > 0$ and $p/n = O(1)$ yields $\|\epsilon_i\|^2 - \mathbb{E}\|\epsilon_i\|^2 = O_p(n^{-1/2})$ as M can be made arbitrary large.

Next, for the cross-term in (A.1), $\epsilon_i^\top \bar{\epsilon} - \mathbb{E}(\epsilon_i^\top \bar{\epsilon})$, we will bound $\text{Var}(\epsilon_i^\top \bar{\epsilon})$ by $\mathbb{E}[(\epsilon_i^\top \bar{\epsilon})^2]$. The first part of the second term can be written as

$$\epsilon_i^\top \bar{\epsilon} = \frac{1}{n} \epsilon_i^\top \epsilon_i + \frac{1}{n} \sum_{k \neq i}^n \epsilon_i^\top \epsilon_k$$

and

$$(\epsilon_i^\top \bar{\epsilon})^2 = \frac{1}{n^2} (\epsilon_i^\top \epsilon_i)^2 + \frac{2}{n} \sum_{k \neq i}^n (\epsilon_i^\top \epsilon_i)(\epsilon_i^\top \epsilon_k) + \frac{1}{n^2} \sum_{k \neq i}^n \sum_{l \neq i}^n (\epsilon_i^\top \epsilon_k)(\epsilon_i^\top \epsilon_l).$$

Since ϵ_i 's are independent and taking expectations, we have the following cases

- for $k \neq i$

$$\begin{aligned} \mathbb{E}[(\epsilon_i^\top \epsilon_i)(\epsilon_i^\top \epsilon_k)] &= \mathbb{E}\{\mathbb{E}[(\epsilon_i^\top \epsilon_i)(\epsilon_i^\top \epsilon_k)] \mid \epsilon_i\} \\ &= \mathbb{E}\{\epsilon_i^\top \epsilon_i \mathbb{E}[\epsilon_i^\top \epsilon_k \mid \epsilon_i]\} \\ &= 0 \quad \text{as} \quad \mathbb{E}[\epsilon_k] = 0. \end{aligned}$$

- for $k \neq l$ and $k, l \neq i$

$$\mathbb{E}[(\epsilon_i^\top \epsilon_k)(\epsilon_i^\top \epsilon_l)] = \mathbb{E}[\epsilon_i^\top \epsilon_k] \mathbb{E}[\epsilon_i^\top \epsilon_l] = 0$$

by independence.

- for $k = l$

$$\mathbb{E}[(\epsilon_i^\top \epsilon_k)(\epsilon_i^\top \epsilon_k)] = \mathbb{E}[(\epsilon_i^\top \epsilon_k)^2].$$

Putting all remaining terms together gives

$$\mathbb{E}[(\epsilon_i^\top \bar{\epsilon})^2] = \frac{1}{n^2} \mathbb{E}[(\epsilon_i^\top \epsilon_i)^2] + \frac{1}{n^2} \sum_{k \neq i}^n \mathbb{E}[(\epsilon_i^\top \epsilon_k)^2]. \quad (\text{A.2})$$

Analyzing the two terms in (A.2), using the variance inequality for sub-Gaussian random variables (Rudelson and Vershynin, 2013) and by Assumption 1

$$\begin{aligned} \mathbb{E}[(\epsilon_i^\top \epsilon_i)^2] &= \text{Var}(\epsilon_i^\top \epsilon_i) + \left\{ \mathbb{E}[(\epsilon_i^\top \epsilon_i)] \right\}^2 \\ &\leq K_\star^4 \text{tr}(\Sigma^2/p^2) + \left\{ \frac{1}{p} \text{tr}(\Sigma) \right\}^2 \\ &= O(1) \end{aligned} \quad (\text{A.3})$$

and

$$\mathbb{E}[(\boldsymbol{\epsilon}_i^\top \boldsymbol{\epsilon}_k)^2] = \mathbb{E}\left\{\mathbb{E}[\boldsymbol{\epsilon}_i^\top (\boldsymbol{\epsilon}_k \boldsymbol{\epsilon}_k^\top) \boldsymbol{\epsilon}_i] \mid \boldsymbol{\epsilon}_i\right\} = \mathbb{E}\left[\boldsymbol{\epsilon}_i^\top \frac{\boldsymbol{\Sigma}}{p} \boldsymbol{\epsilon}_i\right] = \frac{1}{p^2} \text{tr}(\boldsymbol{\Sigma}^2) = O(1/p). \quad (\text{A.4})$$

Substituting (A.3) and (A.4) in (A.2) results in

$$\begin{aligned} \mathbb{E}[(\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}})^2] &= O(n^{-2}) + \frac{n-1}{n^2} O(p^{-1}) \\ &= O\left(\frac{1}{np}\right), \end{aligned}$$

yielding $\text{Var}(\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}}) \leq \mathbb{E}[(\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}})^2] = O\left(\frac{1}{np}\right)$ and hence by Chebyshev's inequality $\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}} - \mathbb{E}(\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}}) = O_p\left(\frac{1}{\sqrt{np}}\right)$.

Finally, for the third term in (A.1), $\|\bar{\boldsymbol{\epsilon}}\|^2 - \mathbb{E}\|\bar{\boldsymbol{\epsilon}}\|^2 = \bar{\boldsymbol{\epsilon}}^\top \bar{\boldsymbol{\epsilon}} - \mathbb{E}(\bar{\boldsymbol{\epsilon}}^\top \bar{\boldsymbol{\epsilon}})$, note that $\bar{\boldsymbol{\epsilon}}$ is a mean zero sub-Gaussian random variable with variance proxy $\boldsymbol{\Sigma}/(np)$. Hence, using the Hanson-Wright inequality with $\mathbf{A} = \mathbf{I}_p$ and $\boldsymbol{\Sigma}_X = \boldsymbol{\Sigma}/(np)$ will give the desired result. Let us bound the Frobenius and operator norms

$$\begin{aligned} \|\mathbf{A} \boldsymbol{\Sigma}_X\|_F^2 &= \frac{1}{n^2 p^2} \text{tr}(\boldsymbol{\Sigma}^2) = O(n^{-2} p^{-1}) \\ \|\mathbf{A} \boldsymbol{\Sigma}_X\| &= \frac{1}{np} \|\boldsymbol{\Sigma}\| = O(n^{-1} p^{-1}). \end{aligned}$$

From Lemma 1, it immediately follows that $\|\bar{\boldsymbol{\epsilon}}\|^2 - \mathbb{E}\|\bar{\boldsymbol{\epsilon}}\|^2 = O_p\left(\frac{1}{\sqrt{np}}\right)$. The result follows by combining the stochastic orders for all three terms.

The second statement of the lemma, $\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j = O_p(n^{-1/2})$, can be shown in a similar manner as the first, and we will therefore only give a summary of the proof. The key idea is to write the quantity $\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j$ as a quadratic form. Define the centering matrix $\mathbf{P} = \mathbf{I}_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top$, hence $\boldsymbol{\omega}_i = \boldsymbol{\epsilon}_i - \bar{\boldsymbol{\epsilon}}$ is the i th p -block of $(\mathbf{P} \otimes \mathbf{I}_p) \mathbf{v}$ where $\mathbf{v} \in \mathbb{R}^{np}$ stacks all vectors $\boldsymbol{\epsilon}_i, i = 1, \dots, n$ and is given by $\mathbf{v} := (\boldsymbol{\epsilon}_1^\top, \dots, \boldsymbol{\epsilon}_n^\top)^\top$. Let $\mathbf{u}_i \in \mathbb{R}^n$ be the i th standard basis vector (i.e., a zero vector of length n with a 1 on the i th position). The p dimensional vector $\boldsymbol{\omega}_i = (\mathbf{u}_i^\top \mathbf{P} \otimes \mathbf{I}_p) \mathbf{v}$ and by the mixed-product property

$$\begin{aligned} \boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j &= \mathbf{v}^\top (\mathbf{u}_i^\top \mathbf{P} \otimes \mathbf{I}_p)^\top (\mathbf{u}_j^\top \mathbf{P} \otimes \mathbf{I}_p) \mathbf{v} \\ &= \mathbf{v}^\top [(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top \otimes \mathbf{I}_p] \mathbf{v}. \end{aligned}$$

The matrix $(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top \otimes \mathbf{I}_p$ is not symmetric, and therefore the Hansen-Wright inequality cannot be directly used. Observe that

$$\mathbf{A} = \frac{1}{2} [(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top + (\mathbf{P} \mathbf{u}_j)(\mathbf{P} \mathbf{u}_i)^\top] \otimes \mathbf{I}_p$$

is symmetric and $\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j = \mathbf{v}^\top \mathbf{A} \mathbf{v}$. The covariance of the stacked vector $\mathbf{v}$ is the block-diagonal matrix $\boldsymbol{\Sigma}_v = \mathbf{I}_n \otimes p^{-1} \boldsymbol{\Sigma}$. Calculating the Frobenius and operator norms yields

$$\begin{aligned}
\|\mathbf{A} \boldsymbol{\Sigma}_v\|_F^2 &= \left\| \left\{ \frac{1}{2} [(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top + (\mathbf{P} \mathbf{u}_j)(\mathbf{P} \mathbf{u}_i)^\top] \otimes \mathbf{I}_p \right\} [\mathbf{I}_n \otimes p^{-1} \boldsymbol{\Sigma}] \right\|_F^2 \\
&= \left\| \frac{1}{2} [(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top + (\mathbf{P} \mathbf{u}_j)(\mathbf{P} \mathbf{u}_i)^\top] \otimes [p^{-1} \mathbf{I}_p \boldsymbol{\Sigma}] \right\|_F^2 \\
&= \left\| \frac{1}{2} [(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top + (\mathbf{P} \mathbf{u}_j)(\mathbf{P} \mathbf{u}_i)^\top] \right\|_F^2 \|p^{-1} \boldsymbol{\Sigma}\|_F^2 \\
&\leq \frac{1}{2} 2 \|\mathbf{P} \mathbf{u}_i\|^2 \|\mathbf{P} \mathbf{u}_j\|^2 \|p^{-1} \boldsymbol{\Sigma}\|_F^2 \quad (\text{bound the symmetric sum of two rank 1 matrices}) \\
&\leq \|p^{-1} \boldsymbol{\Sigma}\|_F^2 \quad (\text{largest eigenvalue of } \mathbf{P} \mathbf{u}_i \text{ is at most 1}) \\
&\leq p^{-2} \|\boldsymbol{\Sigma}\| \text{tr}(\boldsymbol{\Sigma}) = O(1/p),
\end{aligned} \tag{A.5}$$

and

$$\begin{aligned}
\|\mathbf{A} \boldsymbol{\Sigma}_v\| &= \left\| \frac{1}{2} [(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top + (\mathbf{P} \mathbf{u}_j)(\mathbf{P} \mathbf{u}_i)^\top] \otimes p^{-1} \boldsymbol{\Sigma} \right\| \\
&= \left\| \frac{1}{2} [(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top + (\mathbf{P} \mathbf{u}_j)(\mathbf{P} \mathbf{u}_i)^\top] \right\| \|p^{-1} \boldsymbol{\Sigma}\| \\
&\leq \|\mathbf{P} \mathbf{u}_i\| \|\mathbf{P} \mathbf{u}_j\| \|p^{-1} \boldsymbol{\Sigma}\| \\
&= p^{-1} \|\boldsymbol{\Sigma}\| = O(1/p).
\end{aligned} \tag{A.6}$$

Hence, by the Hanson-Wright inequality, we have $\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j - \mathbb{E}[\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j] = O_p(n^{-1/2})$. Next, we need to verify whether $\mathbb{E}[\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j]$ is of lower order than $n^{-1/2}$. Specifically, using the trace properties of matrices, exploiting the structure of the matrix $\mathbf{P}$, then for $i \neq j$

$$\begin{aligned}
\mathbb{E}[\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j] &= \mathbb{E}[\mathbf{v}^\top \mathbf{A} \mathbf{v}] \\
&= \frac{1}{p} \text{tr} \left[\frac{1}{2} [(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top + (\mathbf{P} \mathbf{u}_j)(\mathbf{P} \mathbf{u}_i)^\top] \otimes \boldsymbol{\Sigma} \right] \\
&= \frac{1}{p} \text{tr} [(\mathbf{P} \mathbf{u}_i)(\mathbf{P} \mathbf{u}_j)^\top] \text{tr}(\boldsymbol{\Sigma}) \\
&= (\mathbf{P} \mathbf{u}_j)^\top (\mathbf{P} \mathbf{u}_i) \text{tr}(\boldsymbol{\Sigma}) = \frac{-1}{np} \text{tr}(\boldsymbol{\Sigma}) = O(n^{-1}),
\end{aligned} \tag{A.7}$$

and consequently $\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j = O_p(n^{-1/2})$. ■

Lemma 3 Assume $h = O(1)$, $\gamma = O(1)$ and $\mathbb{E} \mathbf{X}^4 < \infty$. Further, under Assumption 1, Assumption 2 and let $\mathbf{S}^{-1} = (\boldsymbol{\Omega} + \frac{n}{\gamma} \mathbf{I}_n)^{-1}$, then a Neumann series for $\mathbf{S}^{-1}$ converges absolutely in operator norm almost surely under the condition

$$f\left(\frac{\tau}{2h^2}\right) < 1/\gamma$$

for $\tau = \frac{2}{p} \text{tr}(\boldsymbol{\Sigma})$ and kernel matrix $\boldsymbol{\Omega}$ as defined in (2.4).

Proof of Lemma 3. The matrix $\mathbf{S}^{-1}$ can be rewritten as

$$\mathbf{S}^{-1} = \frac{\gamma}{n} \left(\mathbf{I}_n + \frac{\gamma}{n} \mathbf{\Omega} \right)^{-1}.$$

If $\|\mathbf{\Omega}\| < \frac{n}{\gamma}$, then the inverse admits the Neumann series expansion (Horn and Johnson, 2013)

$$\mathbf{S}^{-1} = \left(\mathbf{I}_n + \frac{\gamma}{n} \mathbf{\Omega} \right)^{-1} = \sum_{k=0}^{\infty} (-1)^k \left(\frac{\gamma}{n} \mathbf{\Omega} \right)^k.$$

Next, we will verify when the condition $\|\mathbf{\Omega}\| < \frac{n}{\gamma}$ is valid. From Theorem 2, we have $\frac{1}{p} \|\mathbf{X}_i - \mathbf{X}_j\|^2 \rightarrow \tau^\circ$ a.s. and hence $\tau = \frac{2}{p} \text{tr}(\mathbf{\Sigma}) \rightarrow \tau^\circ$ a.s. Let $\delta_{ij} = \frac{1}{p} \|\mathbf{X}_i - \mathbf{X}_j\|^2 - \tau = O(\sqrt{(\log n)/p})$ a.s., according to Corollary 1. Further, define

$$z_{ij} = \frac{\|\mathbf{X}_i - \mathbf{X}_j\|^2}{2ph^2} = \frac{\delta_{ij}}{2h^2} + \frac{\tau}{2h^2}.$$

A Taylor series for the kernel function of the form $f\left(\frac{\|\mathbf{X}_i - \mathbf{X}_j\|^2}{2ph^2}\right)$ around $z_0 = \frac{\tau}{2h^2}$ is

$$f(z_{ij}) = f(z_0) + f'(z_0)(z_{ij} - z_0) + O_p((z_{ij} - z_0)^2),$$

with

$$z_{ij} - z_0 = \frac{\delta_{ij}}{2h^2} + \frac{\tau}{2h^2} - \frac{\tau}{2h^2} = \frac{\delta_{ij}}{2h^2}$$

and hence, $f'(z_0)(z_{ij} - z_0) = f'(z_0) \frac{\delta_{ij}}{2h^2}$. Set $\Delta_{ij} = \frac{\delta_{ij}}{2h^2}$, then a full series approximation of the kernel matrix $\mathbf{\Omega}$ (or the so-called spike-bulk decomposition) is

$$\mathbf{\Omega} = f(z_0) \mathbf{1}_n \mathbf{1}_n^\top + f'(z_0) \mathbf{\Delta} + \mathbf{R},$$

with $R_{ij} = O(\delta_{ij}^2) = O((\log n)/p)$ a.s. uniformly, assuming $h = O(1)$ and under Assumption 2. Next, we will show that $\|\mathbf{\Delta}\| = O(1)$ a.s. By rewriting δ_{ij} as

$$\delta_{ij} = \|\boldsymbol{\epsilon}_i - \boldsymbol{\epsilon}_j\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_i - \boldsymbol{\epsilon}_j\|^2 = (\|\boldsymbol{\epsilon}_i\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_i\|^2) + (\|\boldsymbol{\epsilon}_j\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_j\|^2) - 2\boldsymbol{\epsilon}_i^\top \boldsymbol{\epsilon}_j$$

and setting $G_{ij} = \boldsymbol{\epsilon}_i^\top \boldsymbol{\epsilon}_j$ for $1 \leq i, j \leq n$ and $D_{ij} = (\|\boldsymbol{\epsilon}_j\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_j\|^2) \mathbf{I}\{i = j\}$, we have

$$2h^2 \mathbf{\Delta} = -2\mathbf{G} + 2\mathbf{D}.$$

It suffices to bound $\|\mathbf{\Delta}\| \leq \frac{1}{2h^2} (2\|\mathbf{G}\| + 2\|\mathbf{D}\|)$. Note that $\mathbf{G}$ can be written as

$$\mathbf{G} = LL^\top \quad \text{with} \quad L = (\boldsymbol{\epsilon}_1^\top, \dots, \boldsymbol{\epsilon}_n^\top)^\top \in \mathbb{R}^{n \times p}$$

and $\|\mathbf{G}\| = \|L\|^2$ where each row of L is mean zero sub-Gaussian with variance proxy $\mathbf{\Sigma}/p$. By Vershynin (2018, Theorem 5.39), we have for $C > 0$

$$\|L\| \leq C \frac{\sqrt{n} + \sqrt{p}}{\sqrt{p}} \|\mathbf{\Sigma}\|^{1/2} \quad \text{a.s.}$$

ans since $p/n \rightarrow \bar{c}_0 \in (0, \infty)$ and $\|\mathbf{\Sigma}\| = O(1)$ it follows that $\|\mathbf{G}\| = O(1)$ *a.s.* The operator norm of $\mathbf{D}$ can be bounded above by the maximum over its diagonal elements in absolute value. By the Hanson-Wright inequality and the union bound

$$\mathbb{P}[\max_i |D_{ii}| > t] \leq 2n \exp(-c'p \min(t^2, t))$$

for a constant $c' > 0$. By taking $t = M > 0$, it follows that $c'p \min(t^2, t) > 2 \log n$. This is always possible since $p/n \rightarrow \bar{c}_0 \in (0, \infty)$ and hence by the Borel-Cantelli lemma $\|\mathbf{D}\| \leq \max_i |D_{ii}| = O(1)$ *a.s.* resulting in $\|\mathbf{\Delta}\| = O(1)$ *a.s.* It remains to show that $\|\mathbf{R}\| = O(1)$ *a.s.* Note that max bounds are too crude, indeed, $\|\mathbf{R}\| \leq \|\mathbf{R}\|_F = O((n \log n)/p)$ *a.s.* and this will diverge under our current data scaling setting.

To obtain a tighter bound for $\|\mathbf{R}\|_F$, we will use moments of δ_{ij} up to order 4. We can use the moment tail integral formula for the moments (Vershynin, 2018, Proposition 2.5.2)

$$\mathbb{E}\delta_{ij}^k = \int_0^\infty kt^{k-1} \mathbb{P}[|\delta_{ij}| > t] dt$$

in combination with the Hanson-Wright inequality with constant $c' > 0$

$$\begin{aligned} \mathbb{E}\delta_{ij}^2 &= \int_0^1 4te^{-c'pt^2} dt + \int_1^\infty 4te^{-c'pt} dt \leq \frac{2}{c'p} + O(e^{-c'p}) = O(1/p) \\ \mathbb{E}\delta_{ij}^4 &= \int_0^1 8t^3e^{-c'pt^2} dt + \int_1^\infty 8t^3e^{-c'pt} dt = O(1/p^2). \end{aligned}$$

We are now ready to bound $\|\mathbf{R}\|_F$. Note that $\mathbb{E}\|\mathbf{R}\|_F^2 = \sum_i \sum_j \mathbb{E}R_{ij}^2 = O(n^2/p^2)$ since $R_{ij}^2 = O(\delta_{ij}^4)$. Since the δ_{ij} 's are not independent and sub-exponential (Wainwright, 2019, Proposition 2.9), we can apply Bernstein's inequality for weakly dependent sub-exponential random variables, such that for all $t > 0$ and a constant $c > 0$ (Merlevède et al., 2009)

$$\mathbb{P}[\|\mathbf{R}\|_F^2 - \mathbb{E}\|\mathbf{R}\|_F^2 > t] \leq 2 \exp\left(-c \min\left(\frac{t^2}{\text{Var}\|\mathbf{R}\|_F^2}, \frac{t}{n^2\|\delta_{ij}^4\|_{\psi_1}}\right)\right)$$

with $\|\delta_{ij}^4\|_{\psi_1}$ is the ψ_1 -Orlicz norm of δ_{ij}^4 . Since δ_{ij}^4 is sub-exponential with scale $1/p^2$ it immediately follows $\|\delta_{ij}^4\|_{\psi_1} \leq c''/p^2$ for some constant $c'' > 0$. It can be shown that $\text{Var}\|\mathbf{R}\|_F^2 = O(n^2/p^2)$; by setting $t = n^2/p^2$ it follows that

$$\mathbb{P}\left[\left|\|\mathbf{R}\|_F^2 - \mathbb{E}\|\mathbf{R}\|_F^2\right| > \frac{n^2}{p^2}\right] \leq 2 \exp(-cn).$$

By Borel-Cantelli, we have $\|\mathbf{R}\|_F^2 \xrightarrow{a.s.} \mathbb{E}\|\mathbf{R}\|_F^2$ and hence by the continuous mapping theorem $\|\mathbf{R}\|_F = O(n/p)$ *a.s.* which equals $O(1)$ under our data scaling assumption.

Combining everything yields

$$\|\mathbf{\Omega}\| = nf(z_0) + O(1) + O(1) < \frac{n}{\gamma} \quad a.s.$$

Consequently, the Neumann series expansion is valid for $\mathbf{S}^{-1}$ if $f(z_0) < 1/\gamma + o(1)$ *a.s.* for $n, p \rightarrow \infty$.

Appendix B.

B. Proof of Theorem 1

First, consider a uniform bound for the term $\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j$ for $i \neq j$. There are $n(n-1) \leq n^2$ pairs for $i \neq j$. Denote the event $E_{ij} = \{|\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j| > t\}$. Then, by Boole's inequality (union bound)

$$\mathbb{P}\left(\bigcup_{i \neq j} E_{ij}\right) \leq \sum_{i \neq j} \mathbb{P}(E_{ij}) \leq n^2 \max_{i \neq j} \mathbb{P}(E_{ij}).$$

Hence, by Lemma 1 in Appendix A, the union bound, Eq. (A.5) and Eq. (A.6), we have

$$\mathbb{P}\left(\max_{i \neq j} |\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j - \mathbb{E}(\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j)| > t\right) \leq 2n^2 \exp \left\{ -c \min \left(\frac{t^2 p^2}{K_\star^4 \text{tr}(\boldsymbol{\Sigma}^2)}, \frac{tp}{K_\star^2 \|\boldsymbol{\Sigma}\|} \right) \right\}.$$

Setting the right hand side equal to $\delta \in (0, 1)$, solving for t and plugging back into the union bound yields the result since $\max_{i \neq j} \mathbb{E}(\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j) = O(n^{-1})$ is negligible compared to $\max_{i \neq j} (\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j)$.

According to the decomposition of ψ_i in Lemma 2, we have

$$\mathbb{P}(|\psi_i| > t) \leq \mathbb{P}\left(|\|\boldsymbol{\epsilon}_i\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_i\|^2| > t/3\right) + \mathbb{P}\left(2|\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}} - \mathbb{E}\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}}| > t/3\right) + \mathbb{P}\left(|\|\bar{\boldsymbol{\epsilon}}\|^2 - \mathbb{E}\|\bar{\boldsymbol{\epsilon}}\|^2| > t/3\right).$$

By using Lemma 1 three times, the union bound (based on n terms), and solving for t , the result follows.

C. Proof of Theorem 2

To write everything in a quadratic form, first, define the stacked np -vector $\boldsymbol{\epsilon} = (\boldsymbol{\epsilon}_1^\top, \dots, \boldsymbol{\epsilon}_n^\top)^\top$. Second, define the $p \times np$ matrix A_i mapping the vector $\boldsymbol{\epsilon}$ to $\boldsymbol{\omega}_i$: for each of the p coordinate directions, the coefficients acting on the n vectors $\{\boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_n\}$ are

$$\left(\underbrace{\frac{n-1}{n}}_{\text{acting on } \boldsymbol{\epsilon}_i}, \underbrace{\frac{-1}{n}, \dots, \frac{-1}{n}}_{\text{for } k \neq i} \right).$$

Consequently,

$$\boldsymbol{\omega}_i = A_i \boldsymbol{\epsilon} \quad \text{and} \quad \psi_i = \|A_i \boldsymbol{\epsilon}\|^2 - \mathbb{E}\|A_i \boldsymbol{\epsilon}\|^2 = \boldsymbol{\epsilon}^\top M_i \boldsymbol{\epsilon} - \mathbb{E}(\boldsymbol{\epsilon}^\top M_i \boldsymbol{\epsilon}),$$

with $M_i = A_i^\top A_i$. Similarly, we have

$$\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j = \boldsymbol{\epsilon}^\top B_{ij} \boldsymbol{\epsilon} \quad \text{with} \quad B_{ij} = A_i^\top A_j.$$

Now, all three random terms $(\psi_i, \psi_j, \boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j)$ are centered quadratic forms in the stacked vector $\boldsymbol{\epsilon}$.

To use the Hanson-Wright inequality, we need to bound the operator and Frobenius norms of A_i , M_i , and B_{ij} . For any of the p th fixed coordinate, the coefficient vector of length

n used by A_i is $((n-1/n), -1/n, \dots, -1/n)$ has Euclidean norm equal to $\sqrt{(n-1)/n}$ and hence the operator and Frobenius norms are

$$\|A_i\| = \sqrt{\frac{n-1}{n}} \quad \text{and} \quad \|A_i\|_F = \sqrt{p} \sqrt{\frac{n-1}{n}},$$

and

$$\|M_i\| = \|A_i\|^2 = \frac{n-1}{n} \quad \text{and} \quad \|M_i\|_F \leq \|A_i\| \|A_i\|_F = \sqrt{p} \frac{n-1}{n}.$$

For the cross matrix $B_{ij} = A_i^\top A_j$, we have

$$\|B_{ij}\| \leq \|A_i\| \|A_j\| = \frac{n-1}{n} \quad \text{and} \quad \|B_{ij}\|_F \leq \|A_i\|_F \|A_j\|_F = p \frac{n-1}{n}.$$

By the Hanson-Wright inequality (see Lemma 1) for $t > 0$,

$$\begin{aligned} \mathbb{P}(|\psi_i| > t) &\leq 2 \exp \left(-c \min \left\{ \frac{t^2}{K_\star^4 (\sqrt{p} \frac{n-1}{n})^2}, \frac{t}{K_\star^2 \frac{n-1}{n}} \right\} \right) \\ &= 2 \exp \left(-c \min \left\{ \frac{n^2 t^2}{K_\star^4 p (n-1)^2}, \frac{nt}{K_\star^2 (n-1)} \right\} \right) \end{aligned}$$

and

$$\mathbb{P}(|\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j - \mathbb{E}[\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j]|) \leq 2 \exp \left(-c \min \left\{ \frac{n^2 t^2}{K_\star^4 p^2 (n-1)^2}, \frac{nt}{K_\star^2 (n-1)} \right\} \right).$$

For completeness, we have included the term $\mathbb{E}[\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j]$ in the above inequality. However, from (A.7) we have $\mathbb{E}[\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j] = O(n^{-1})$ and is negligible.

Next, define $\Delta_{ij}(p) := \frac{1}{p} \|\mathbf{X}_j - \mathbf{X}_i\|^2 - \tau^\circ = \psi_i + \psi_j - 2\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j$. We can now obtain a bound for $\mathbb{P}(|\Delta_{ij}(p)| > u)$. Fix $u > 0$ and for $i \neq j$, then by the triangle inequality

$$\{|\Delta_{ij}(p)| > u\} \subseteq \left\{ |\psi_i| > \frac{pu}{3} \right\} \cup \left\{ |\psi_j| > \frac{pu}{3} \right\} \cup \left\{ 2|\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j| > \frac{pu}{3} \right\}.$$

Applying union bound and the Hanson-Wright probability bounds above, then for the constant $c_1 > 0$ (depending only on c , and containing the numeric factors $1/9$ from the $pu/3$ split), such that for every fixed $u > 0$, $i \neq j$, and every p

$$\mathbb{P} \left(\left| \frac{1}{p} \|\mathbf{X}_j - \mathbf{X}_i\|^2 - \tau^\circ \right| > u \right) \leq 6 \exp \left(-c_1 \min \left\{ \frac{n^2 u^2}{K_\star^4 p (n-1)^2}, \frac{np u}{K_\star^2 (n-1)} \right\} \right). \quad (\text{C.1})$$

The most important feature is that for fixed $u > 0$, the right-hand side decays at least like $\exp(-c_2 p)$ for large p where $c_2 > 0$ is constant depending on $K_\star, n$, bound on $\|\boldsymbol{\Sigma}\|$ but not on p . Indeed, for any fixed $u > 0$, the minimum is eventually attained by the second argument, so there exists a $p_0(u)$ and $c' > 0$ such that

$$\mathbb{P}(|\Delta_{ij}(p)| > u) \leq 6 \exp(-c' p), \quad \text{for all } p \geq p_0(u). \quad (\text{C.2})$$

Summing the tail probabilities over $p \in \mathbb{N}$ and for fixed $u > 0$ results in

$$\sum_{p=1}^{\infty} \mathbb{P}(|\Delta_{ij}(p)| > u) \leq \sum_{p=1}^{p_0(u)-1} \mathbb{P}(|\Delta_{ij}(p)| > u) + \sum_{p=p_0(u)}^{\infty} 6e^{-c' p}.$$

The first argument is a finite sum of probabilities, and the second argument is a geometric series, which is finite. Consequently, the whole series is finite, yielding $\sum_{p=1}^{\infty} \mathbb{P}(|\Delta_{ij}(p)| > u) < \infty$. According to the Borel-Cantelli lemma, with probability 1, only finitely many events $\{|\Delta_{ij}(p)| > u\}$ occur. Equivalently,

$$\mathbb{P}(\limsup_{p \rightarrow \infty} |\Delta_{ij}(p)| > u) = 0,$$

and hence $|\Delta_{ij}(p)| \rightarrow 0$ a.s. for that specific fixed $u > 0$. To complete the proof, we need to take a sequence $u_m \downarrow 0$, for example $u_m = 1/m$. By the countable union of measure-zero sets, with probability 1, $|\Delta_{ij}(p)| \rightarrow 0$ a.s. for all u_m and hence $|\Delta_{ij}(p)| \rightarrow 0$ a.s.

For the second part of the proof, note that there are $\binom{n}{2} \leq n^2$ distinct pairs (i, j) . By (C.1), (C.2) and the union bound, we have for $i \neq j$

$$\mathbb{P}(\max_{i \neq j} |\Delta_{ij}(p)| > u) \leq Cn^2 e^{-c'p}, \quad (\text{C.3})$$

for constants $C, c' > 0$ independent of p . It immediately follows that

$$\sum_{p=1}^{\infty} \mathbb{P}(\max_{i \neq j} |\Delta_{ij}(p)| > u) < \infty,$$

and by Borel-Cantelli

$$\max_{i \neq j} \left| \frac{1}{p} \|\mathbf{X}_j - \mathbf{X}_i\| - \tau^\circ \right| \xrightarrow{a.s.} 0.$$

Furthermore, notice that this result also holds under our assumption of $p = O(n)$, by simply substituting $p = c_3 n$ in (C.3), where c_3 is a constant greater than 0, independent of p .

The second part of the theorem is shown by using Corollary 1 and by noting that $\max_{i \neq j} \left| \frac{1}{p} \|\mathbf{X}_i - \mathbf{X}_j\|^2 - \tau^\circ \right| \leq \max_i |\psi_i| + \max_j |\psi_j| + 2 \max_{i \neq j} |\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j|$.

D. Proof of Theorem 3

Let $\mathbf{X}_i = \boldsymbol{\mu} + \sqrt{p}\boldsymbol{\epsilon}_i$, where $\boldsymbol{\epsilon}_i \stackrel{\text{ind}}{\sim} \text{subG}(p^{-1}\boldsymbol{\Sigma})$, where p is the dimension of $\mathbf{X}$ and let $\boldsymbol{\omega}_i = \boldsymbol{\epsilon}_i - \bar{\boldsymbol{\epsilon}}$. By expanding the Euclidean norm

$$\frac{1}{p} \|\mathbf{X}_j - \mathbf{X}_i\|^2 = \frac{1}{p} \|\boldsymbol{\mu} + \sqrt{p}\boldsymbol{\epsilon}_j - \boldsymbol{\mu} - \sqrt{p}\boldsymbol{\epsilon}_i\|^2 = \|\boldsymbol{\epsilon}_j - \boldsymbol{\epsilon}_i\|^2$$

and

$$\begin{aligned} \|\boldsymbol{\epsilon}_j - \boldsymbol{\epsilon}_i\|^2 &= \|\boldsymbol{\epsilon}_j - \bar{\boldsymbol{\epsilon}} - \boldsymbol{\epsilon}_i + \bar{\boldsymbol{\epsilon}}\|^2 = \|\boldsymbol{\omega}_j - \boldsymbol{\omega}_i\|^2 \\ &= \|\boldsymbol{\omega}_j\|^2 - \mathbb{E} \|\boldsymbol{\omega}_j\|^2 + \|\boldsymbol{\omega}_i\|^2 - \mathbb{E} \|\boldsymbol{\omega}_i\|^2 - 2\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j + 2\mathbb{E} \|\boldsymbol{\omega}_j\|^2 \\ &= \psi_j + \psi_i - 2\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j + \tau^\circ \end{aligned} \quad (\text{D.1})$$

with $\psi_i = \|\boldsymbol{\omega}_i\|^2 - \mathbb{E} \|\boldsymbol{\omega}_i\|^2$ and $\tau^\circ = 2\mathbb{E} \|\boldsymbol{\omega}_j\|^2$. The first two terms in (D.1) capture the fluctuation of the squared norm of $\boldsymbol{\omega}_i$ and $\boldsymbol{\omega}_j$ respectively, the third term captures the

correlations between deviations, and the last term is a deterministic baseline term of the squared distance. Using the expectation of quadratic forms and $\mathbb{E}(\boldsymbol{\epsilon}_i \boldsymbol{\epsilon}_i^\top) = p^{-1} \boldsymbol{\Sigma}$

$$\begin{aligned}
\mathbb{E} \|\boldsymbol{\omega}_i\|^2 &= \mathbb{E} \|\boldsymbol{\epsilon}_i - \bar{\boldsymbol{\epsilon}}\|^2 \\
&= \mathbb{E} \|\boldsymbol{\epsilon}_i\|^2 - 2\mathbb{E}(\boldsymbol{\epsilon}_i^\top \bar{\boldsymbol{\epsilon}}) + \mathbb{E} \|\bar{\boldsymbol{\epsilon}}\|^2 \\
&= \mathbb{E} \|\boldsymbol{\epsilon}_i\|^2 - \frac{2}{n} \mathbb{E} \|\boldsymbol{\epsilon}_i\|^2 + \frac{1}{n} \mathbb{E} \|\boldsymbol{\epsilon}_i\|^2 \\
&= \frac{1}{p} \text{tr}(\boldsymbol{\Sigma}) - \frac{1}{np} \text{tr}(\boldsymbol{\Sigma}) \\
&= \frac{n-1}{np} \text{tr}(\boldsymbol{\Sigma})
\end{aligned}$$

and consequently, $\tau^\circ = 2\mathbb{E} \|\boldsymbol{\omega}_i\|^2 = \frac{2}{p} \frac{n-1}{n} \text{tr}(\boldsymbol{\Sigma}) = \tau + O(n^{-1})$ a.s. since $\text{tr}(\boldsymbol{\Sigma}) = O(p)$, see covariance scaling in Assumption 1. A Taylor expansion around τ° for $f(\frac{1}{p} \|\mathbf{X}_j - \mathbf{X}_i\|^2)$ is

$$\begin{aligned}
f\left(\frac{1}{p} \|\mathbf{X}_j - \mathbf{X}_i\|^2\right) &= f(\tau^\circ) + f'(\tau^\circ)(\psi_j + \psi_i - 2\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j) + \\
&\quad \frac{f''(\tau^\circ)}{2}(\psi_j^2 + \psi_i^2 + 2\psi_i \psi_j^\top + \frac{4}{p^2} \|\boldsymbol{\Sigma}\|_F^2) + R.
\end{aligned} \tag{D.2}$$

The order of the remainder term $R = O\left(\left|\frac{1}{p} \|\mathbf{X}_j - \mathbf{X}_i\|^2 - \tau^\circ\right|^3\right)$ of the Taylor series is $O(n^{-3/2})$ a.s. This can be easily derived using (C.1) with $u = Mn^{-1/2}$, $M > 0$ and applying Borell-Cantelli. Eq. (D.2) is valid for all off-diagonal entries of K and all diagonal elements are $f(0)$. Following El Karoui (2010), we can write the kernel matrix as

$$\begin{aligned}
\boldsymbol{\Omega} &= f(\tau^\circ) \mathbf{1}_n \mathbf{1}_n^\top + f'(\tau^\circ)(\boldsymbol{\psi} \mathbf{1}_n^\top + \mathbf{1}_n \boldsymbol{\psi}^\top - 2\mathbf{P} \mathbf{W}^\top \mathbf{W} \mathbf{P}) + \\
&\quad \frac{f''(\tau^\circ)}{2!}(\boldsymbol{\psi}^2 \mathbf{1}_n^\top + \mathbf{1}_n \boldsymbol{\psi}^2^\top + 2\boldsymbol{\psi} \boldsymbol{\psi}^\top + \frac{4}{p^2} \|\boldsymbol{\Sigma}\|_F^2 \mathbf{1}_n \mathbf{1}_n^\top) + \\
&\quad (f(0) - f(\tau^\circ) + \tau f'(\tau^\circ)) \mathbf{I}_n + O(n^{-1/2}) \text{ a.s.} \\
&= -2f'(\tau^\circ)(\mathbf{P} \mathbf{W}^\top \mathbf{W} \mathbf{P} + \mathbf{A}) + \beta \mathbf{I}_n + O(n^{-1/2}) \text{ a.s.}
\end{aligned} \tag{D.3}$$

where

$$\begin{aligned}
\beta &= f(0) - f(\tau^\circ) + \tau^\circ f'(\tau^\circ) \\
\mathbf{A} &= \mathbf{A}_n + \mathbf{A}_{\sqrt{n}} + \mathbf{A}_1
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{A}_n &= -\frac{f(\tau^\circ)}{2f'(\tau^\circ)} \mathbf{1}_n \mathbf{1}_n^\top \\
\mathbf{A}_{\sqrt{n}} &= -\frac{1}{2}(\boldsymbol{\psi} \mathbf{1}_n^\top + \mathbf{1}_n \boldsymbol{\psi}^\top) \\
\mathbf{A}_1 &= -\frac{f''(\tau^\circ)}{4f'(\tau^\circ)}(\boldsymbol{\psi}^2 \mathbf{1}_n^\top + \mathbf{1}_n \boldsymbol{\psi}^2^\top + \frac{4}{p^2} \|\boldsymbol{\Sigma}\|_F^2 \mathbf{1}_n \mathbf{1}_n^\top + 2\boldsymbol{\psi} \boldsymbol{\psi}^\top)
\end{aligned}$$

Consequently, an approximation to the kernel matrix $\boldsymbol{\Omega}$ is

$$\tilde{\boldsymbol{\Omega}} = -2f'(\tau)(\mathbf{P} \mathbf{W}^\top \mathbf{W} \mathbf{P} + \mathbf{A}) + \beta \mathbf{I}_n.$$

E. High-dimensional covariance estimation

Estimating covariance matrices in high dimensions remains a central challenge in statistics, as the classical sample covariance estimator is no longer reliable when the dimension (p) is comparable to or exceeds the sample size (n) (Pourahmadi, 2013). In this regime, the sample covariance matrix is singular for ($p \geq n$) and exhibits severe eigenvalue distortion even when invertible, with large eigenvalues inflated and small ones deflated due to noise rather than signal (Johnstone, 2001). From a random matrix perspective, the empirical spectrum spreads out and deviates substantially from the population spectrum, obscuring the underlying low-rank or structured components that often drive dependence. These issues arise because the estimator attempts to estimate $O(p^2)$ parameters without any structural constraints, leading to inconsistency and instability in downstream tasks such as principal component analysis, discriminant analysis, portfolio optimization, and risk management. Moreover, the inverse covariance (precision) matrix—central to regression adjustment and graphical modeling—becomes ill-conditioned or undefined, further amplifying estimation error. As a result, naive plug-in procedures based on the sample covariance matrix can perform arbitrarily poorly in high-dimensional settings.

Modern high-dimensional covariance estimation methods address these shortcomings by explicitly reducing effective complexity through regularization or structural assumptions. Shrinkage estimators, including Stein-type and oracle-based approaches, improve conditioning and reduce variance by shrinking the sample covariance toward a low-complexity target such as a multiple of the identity or a factor-structured matrix (Ledoit and Wolf, 2004; Zhang et al., 2023). These approaches balance bias and variance in a controlled manner and often admit optimality properties under quadratic loss. Sparsity-based methods assume that many covariances (or partial correlations) are negligible and employ thresholding or penalization to achieve consistency under appropriate sparsity conditions (Bickel and Levina, 2008a; Wang et al., 2023). Such estimators can attain favorable rates of convergence under operator or Frobenius norms and are particularly effective when dependence is localized.

Banding and tapering techniques further exploit natural orderings of variables—common in time series, longitudinal, or spatial data—by assuming dependence decays with distance, either sharply through banding or smoothly through tapering, yielding estimators with improved convergence rates and reduced variance (Bickel and Levina, 2008b; Cai et al., 2010). Graphical model approaches instead impose sparsity on the precision matrix, enabling interpretable conditional independence structures and scalable computation via convex optimization (Friedman et al., 2008). More recently, factor-based, structured, and Bayesian methods have been developed to handle ultra-high dimensions, small sample sizes, and evolving covariance structures, incorporating latent factors, hierarchical priors, or dynamic components to capture complex dependence patterns (Fan et al., 2013; Puchkin and Rakhuba, 2024). Many of these methods provide non-asymptotic error bounds, adaptivity to unknown structure, and robustness to model misspecification. Collectively, this body of work demonstrates that successful covariance estimation in high dimensions requires moving beyond classical estimators and adopting principled regularization strategies that explicitly account for dimensionality and structural complexity—an extensive literature that lies beyond the scope of this paper.

F. Proof of Theorem 4

From Theorem 3, we know that $\mathbf{\Omega} = \tilde{\mathbf{\Omega}} + \mathbf{R}_n$, where the matrix $\mathbf{R}_n$ comprises the higher order terms of the Taylor expansion. It suffices to show that $\|\mathbf{R}_n\| \rightarrow 0$ a.s. Working component-wise, the Lagrange form of the remainder for i and j is

$$R_{n,ij} = \frac{f^{(3)}(\tau_{ij}^\circ)}{6}(\psi_i + \psi_j - 2\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j)^3$$

for some real τ_{ij}° between τ° and $\psi_i + \psi_j - 2\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j$. By Corollary 1, the C_r inequality and Assumption 2, we have that

$$\begin{aligned} \max_{i,j} |R_{n,ij}| &\leq \frac{9}{6} \sup_x |f^{(3)}(x)| \{ (\max_i |\psi_i|)^3 + (\max_j |\psi_j|)^3 + 8(\max_{i \neq j} |\boldsymbol{\omega}_i^\top \boldsymbol{\omega}_j|)^3 \} \\ &= O((\log n)^{3/2} p^{-3/2}) \quad a.s. \end{aligned}$$

We can bound the operator norm of the $n \times n$ matrix $\mathbf{R}_n$ by

$$\|\mathbf{R}_n\| \leq n \max_{i,j} |R_{n,ij}| \rightarrow 0 \quad a.s.$$

under Assumption 1 and 2.

G. Proof of Theorem 5

First, we will approximate the matrix $\mathbf{S}^{-1} = (\mathbf{\Omega} + \frac{n}{\gamma} \mathbf{I}_n)^{-1}$ with

$$\mathbf{S}^{-1} = \frac{1}{n} \left[\frac{1}{\gamma} \mathbf{I}_n + \frac{1}{n} \tilde{\mathbf{\Omega}} + O(n^{-3/2}) \right]^{-1} \quad a.s.,$$

by Theorem 3. Using Eq. (D.3) and identify the leading order terms of $\tilde{\mathbf{\Omega}}$ i.e., $-2f'(\tau^\circ) \mathbf{A}_n = f(\tau^\circ) \mathbf{1}_n \mathbf{1}_n^\top$ and $\frac{n}{\gamma} \mathbf{I}_n$ (both operator norms are $O(n)$ a.s.) yields

$$\mathbf{S}^{-1} = \frac{1}{n} \left[\frac{\mathbf{I}_n}{\gamma} + \frac{f(\tau^\circ) \mathbf{1}_n \mathbf{1}_n^\top}{n} + \mathbf{E}_n \right]^{-1}$$

where $\mathbf{E}_n = -\frac{2f'(\tau^\circ)}{n} (\mathbf{P} \mathbf{W}^\top \mathbf{W} \mathbf{P} + \mathbf{A}_{\sqrt{n}} + \mathbf{A}_1) + \frac{\beta}{n} \mathbf{I}_n + O(n^{-3/2})$ a.s. Define $\mathbf{L}^{-1} = \frac{1}{\gamma} \mathbf{I}_n + \frac{f(\tau^\circ)}{n} \mathbf{1}_n \mathbf{1}_n^\top = O(1)$ a.s. then

$$\mathbf{S}^{-1} = \frac{1}{n} [\mathbf{L}^{-1} + \mathbf{E}_n]^{-1} \quad \text{and} \quad \|\mathbf{E}_n\| = O(n^{-1}) \quad a.s.$$

Based on Lemma 3, $\|\mathbf{\Omega}\| < n/\gamma$ a.s. and hence by Theorem 4 we also have $\|\tilde{\mathbf{\Omega}}\| < n/\gamma$ a.s.. This implies that the Neumann series converges absolutely in operator norm almost surely (see e.g Rudin (1991) and Vershynin (2018))

$$\begin{aligned} \mathbf{S}^{-1} &= \frac{1}{n} \left[\mathbf{L} - \mathbf{L} \mathbf{E}_n \mathbf{L} + \mathbf{L} \mathbf{E}_n \mathbf{L} \mathbf{E}_n \mathbf{L} + O(n^{-3/2}) \right] a.s. \\ &= \frac{\mathbf{L}}{n} - \frac{\mathbf{L} \mathbf{E}_n \mathbf{L}}{n} + \frac{\mathbf{L} \mathbf{E}_n \mathbf{L} \mathbf{E}_n \mathbf{L}}{n} + O(n^{-5/2}) \quad a.s. \end{aligned} \tag{G.1}$$

For the second term, we have

$$-\frac{\mathbf{L}\mathbf{E}_n\mathbf{L}}{n} = \frac{2f'(\tau^\circ)}{n^2}\mathbf{L}\mathbf{A}_{\sqrt{n}}\mathbf{L} - \mathbf{L}\frac{\beta}{n^2}\mathbf{L} + \frac{2f'(\tau^\circ)}{n^2}\mathbf{L}\left(\mathbf{A}_1 + \mathbf{P}\mathbf{W}^\top\mathbf{W}\mathbf{P}\right)\mathbf{L} + O(n^{-5/2}) a.s.$$

and for the third term, while keeping orders only to $O(n^{-2})$

$$\frac{\mathbf{L}\mathbf{E}_n\mathbf{L}\mathbf{E}_n\mathbf{L}}{n} = \frac{4(f'(\tau^\circ))^2}{n^3}\mathbf{L}\mathbf{A}_{\sqrt{n}}\mathbf{L}\mathbf{A}_{\sqrt{n}}\mathbf{L} + O(n^{-3}) a.s.$$

Combining the results for the second and third term in (G.1) yields

$$\begin{aligned}\mathbf{S}^{-1} &= \frac{\mathbf{L}}{n} + \frac{2f'(\tau^\circ)}{n^2}\mathbf{L}\mathbf{A}_{\sqrt{n}}\mathbf{L} + \mathbf{L}\left(\mathbf{Q} - \frac{\beta}{n^2}\mathbf{I}_n\right)\mathbf{L} + O(n^{-5/2}) a.s. \\ &= \mathcal{S}_{(1,n^{-1})} + \mathcal{S}_{(1,n^{-3/2})} + \mathcal{S}_{(1,n^{-2})} + O(n^{-5/2}) a.s.,\end{aligned}$$

with

$$\mathbf{Q} = \frac{2f'(\tau^\circ)}{n^2}\left(\mathbf{A}_1 + \mathbf{P}\mathbf{W}^\top\mathbf{W}\mathbf{P} + \frac{2f'(\tau^\circ)}{n}\mathbf{A}_{\sqrt{n}}\mathbf{L}\mathbf{A}_{\sqrt{n}}\right).$$

Using the Sherman-Morrison formula (Horn and Johnson, 2013, p. 19, §0.7.4), we have almost surely

$$\mathbf{L} = \left(\frac{1}{\gamma}\mathbf{I}_n + \frac{f(\tau^\circ)}{n}\mathbf{1}_n\mathbf{1}_n^\top\right)^{-1} = \frac{\gamma}{1 + \gamma f(\tau^\circ)}\mathbf{I}_n + \frac{\gamma^2 f(\tau^\circ)}{1 + \gamma f(\tau^\circ)}\mathbf{P} = O(1)$$

Also, $\mathbf{P} = \mathbf{I}_n - \frac{\mathbf{1}_n\mathbf{1}_n^\top}{n}$ and $\mathbf{1}_n^\top\mathbf{P} = \mathbf{P}\mathbf{1}_n = 0$. This implies $\mathbf{L}\mathbf{1}_n = \frac{\gamma}{1 + \gamma f(\tau)}\mathbf{1}_n$ and hence $\mathbf{L} - \frac{\mathbf{1}_n\mathbf{1}_n^\top\mathbf{L}}{n} = \gamma\mathbf{P}$. Now, we compute $\mathbf{1}_n^\top\mathbf{S}^{-1}$ as it is an important quantity for the approximation of $\hat{\alpha}$ and $\hat{b}$.

$$\begin{aligned}\mathbf{1}_n^\top\mathbf{S}^{-1} &= \frac{\mathbf{1}_n^\top\mathbf{L}}{n} + \frac{2f'(\tau^\circ)}{n^2}\mathbf{1}_n^\top\mathbf{L}\mathbf{A}_{\sqrt{n}}\mathbf{L} + \mathbf{1}_n^\top\mathbf{L}\left(\mathbf{Q} - \frac{\beta}{n^2}\mathbf{I}_n\right)\mathbf{L} + O(n^{-3/2}) a.s. \\ &= \mathcal{R}_{(1,1)} + \mathcal{R}_{(1,n^{-1/2})} + \mathcal{R}_{(1,n^{-1})} + O(n^{-3/2}) a.s.\end{aligned}\tag{G.2}$$

Using (G.2), the approximation for the quantities $\mathbf{1}_n\mathbf{1}_n^\top\mathbf{S}^{-1}$, $\mathbf{1}_n^\top\mathbf{S}^{-1}\mathbf{Y}$, and $\mathbf{1}_n^\top\mathbf{S}^{-1}\mathbf{1}_n$ is

$$\begin{aligned}\mathbf{1}_n^\top\mathbf{S}^{-1}\mathbf{Y} &= \frac{\mathbf{1}_n^\top\mathbf{L}\mathbf{Y}}{n} + \frac{2f'(\tau^\circ)}{n^2}\mathbf{1}_n^\top\mathbf{L}\mathbf{A}_{\sqrt{n}}\mathbf{L}\mathbf{Y} + \mathbf{1}_n^\top\mathbf{L}\left(\mathbf{Q} - \frac{\beta}{n^2}\mathbf{I}_n\right)\mathbf{L}\mathbf{Y} + O(n^{-3/2}) a.s. \\ &= \mathcal{R}_{(2,1)} + \mathcal{R}_{(2,n^{-1/2})} + \mathcal{R}_{(2,n^{-1})} + O(n^{-3/2}) a.s.,\end{aligned}\tag{G.3}$$

$$\begin{aligned}\mathbf{1}_n^\top\mathbf{S}^{-1}\mathbf{1}_n &= \frac{\mathbf{1}_n^\top\mathbf{L}\mathbf{1}_n}{n} + \frac{2f'(\tau^\circ)}{n^2}\mathbf{1}_n^\top\mathbf{L}\mathbf{A}_{\sqrt{n}}\mathbf{L}\mathbf{1}_n + \mathbf{1}_n^\top\mathbf{L}\left(\mathbf{Q} - \frac{\beta}{n^2}\mathbf{I}_n\right)\mathbf{L}\mathbf{1}_n + O(n^{-3/2}) a.s., \\ &= \mathcal{R}_{(3,1)} + \mathcal{R}_{(3,n^{-1/2})} + \mathcal{R}_{(3,n^{-1})} + O(n^{-3/2}) a.s.,\end{aligned}\tag{G.4}$$

$$\begin{aligned}\mathbf{1}_n\mathbf{1}_n^\top\mathbf{S}^{-1} &= \frac{\mathbf{1}_n\mathbf{1}_n^\top\mathbf{L}}{n} + \frac{2f'(\tau^\circ)}{n^2}\mathbf{1}_n\mathbf{1}_n^\top\mathbf{L}\mathbf{A}_{\sqrt{n}}\mathbf{L} + \mathbf{1}_n\mathbf{1}_n^\top\mathbf{L}\left(\mathbf{Q} - \frac{\beta}{n^2}\mathbf{I}_n\right)\mathbf{L} + O(n^{-3/2}) a.s. \\ &= \mathcal{R}_{(4,1)} + \mathcal{R}_{(4,n^{-1/2})} + \mathcal{R}_{(4,n^{-1})} + O(n^{-3/2}) a.s.\end{aligned}$$

Next, we approximate $(\mathbf{1}_n^\top \mathbf{S}^{-1} \mathbf{1}_n)^{-1}$ by a Neumann series around its leading order and using Lemma 3, allowing for an error term of $O(n^{-3/2})$ a.s.

$$\begin{aligned} (\mathbf{1}_n^\top \mathbf{S}^{-1} \mathbf{1}_n)^{-1} &= \frac{1 + \gamma f(\tau^\circ)}{\gamma} \left[1 - \frac{2f'(\tau^\circ)}{n^2} \frac{\gamma \mathbf{1}_n^\top \mathbf{A}_{\sqrt{n}} \mathbf{1}_n}{1 + \gamma f(\tau^\circ)} - \frac{\gamma \mathbf{1}_n^\top \left(\mathbf{Q} - \frac{\beta}{n^2} \mathbf{I}_n \right) \mathbf{1}_n}{1 + \gamma f(\tau^\circ)} \right] + O(n^{-3/2}) \text{ a.s.} \\ &= \mathcal{R}_{(5,1)} + \mathcal{R}_{(5,n^{-1/2})} + \mathcal{R}_{(5,n^{-1})} + O(n^{-3/2}) \text{ a.s.} \end{aligned}$$

Multiplying (G.3) and (G.4), the bias term $\hat{b}$ is approximated by

$$\begin{aligned} \hat{b} &= \frac{\mathbf{1}_n^\top \mathbf{S}^{-1} \mathbf{Y}}{\mathbf{1}_n^\top \mathbf{S}^{-1} \mathbf{1}_n} \\ &= \mathcal{R}_{(2,1)} \mathcal{R}_{(5,1)} + \mathcal{R}_{(5,1)} \mathcal{R}_{(2,n^{-1/2})} + \mathcal{R}_{(2,1)} \mathcal{R}_{(5,n^{-1/2})} + \mathcal{R}_{(5,1)} \mathcal{R}_{(2,n^{-1})} + \mathcal{R}_{(2,1)} \mathcal{R}_{(5,n^{-1})} \\ &\quad + \mathcal{R}_{(2,n^{-1/2})} \mathcal{R}_{(5,n^{-1/2})} + O(n^{-3/2}) \text{ a.s.} \\ &= \mathcal{B}_{(1,1)} + \mathcal{B}_{(1,n^{-1/2})} + \mathcal{B}_{(1,n^{-1})} + O(n^{-3/2}) \text{ a.s.}, \end{aligned} \tag{G.5}$$

with $\mathcal{B}_{(1,1)} = \bar{Y}$, $\mathcal{B}_{(1,n^{-1/2})} = -\frac{\gamma f'(\tau^\circ)}{n} \boldsymbol{\psi}^\top \mathbf{P} \mathbf{Y}$, and $\mathcal{B}_{(1,n^{-1})} = -\frac{\gamma f''(\tau^\circ)}{2n} \boldsymbol{\psi}^2 \mathbf{P} \mathbf{Y}$. Finally, (G.5) can be written as

$$\hat{b} = \bar{Y} - \frac{\gamma f'(\tau^\circ)}{n} \boldsymbol{\psi}^\top \mathbf{P} \mathbf{Y} - \frac{\gamma f''(\tau^\circ)}{2n} \boldsymbol{\psi}^2 \mathbf{P} \mathbf{Y} + O(n^{-3/2}) \text{ a.s.} \tag{G.6}$$

Similarly,

$$\begin{aligned} \hat{\alpha} &= \mathbf{S}^{-1} \left(\mathbf{I}_n - \frac{\mathbf{1}_n \mathbf{1}_n^\top \mathbf{S}^{-1}}{\mathbf{1}_n^\top \mathbf{S}^{-1} \mathbf{1}_n} \right) \mathbf{Y} \\ &= \mathcal{S}_{(1,n^{-1})} \mathbf{Y} - \mathcal{S}_{(1,n^{-1})} \mathbf{1}_n \mathcal{B}_{(1,1)} + \mathcal{S}_{(1,n^{-3/2})} \mathbf{Y} - \mathcal{S}_{(1,n^{-1})} \mathbf{1}_n \mathcal{B}_{(1,n^{-1/2})} - \mathcal{S}_{(1,n^{-3/2})} \mathbf{1}_n \mathcal{B}_{(1,n^{-1/2})} \\ &\quad - \mathcal{S}_{(1,n^{-2})} \mathbf{1}_n \mathcal{B}_{(1,1)} + O(n^{-5/2}) \text{ a.s.} \\ &= \mathcal{B}_{(2,n^{-1})} + \mathcal{B}_{(2,n^{-3/2})} + \mathcal{B}_{(2,n^{-2})} + O(n^{-5/2}) \text{ a.s.}, \end{aligned}$$

with $\mathcal{B}_{(2,n^{-1})} = \frac{\gamma}{n} \mathbf{P} \mathbf{Y}$, $\mathcal{B}_{(2,n^{-3/2})} = 0$ and $\mathcal{B}_{(2,n^{-2})} = \gamma \mathbf{P} \left(\mathbf{Q} - \frac{\beta}{n^2} \mathbf{I}_n \right) \mathbf{P} \mathbf{Y} - \frac{\gamma^3 (f'(\tau^\circ))^2}{n^2 (1 + \gamma f(\tau^\circ))} \boldsymbol{\psi} \boldsymbol{\psi}^\top \mathbf{P} \mathbf{Y}$ since $\mathbf{P} \mathbf{A}_{\sqrt{n}} \mathbf{P} = 0$ and consequently,

$$\hat{\alpha} = \frac{\gamma}{n} \mathbf{P} \mathbf{Y} + \gamma \mathbf{P} \left(\mathbf{Q} - \frac{\beta}{n^2} \mathbf{I}_n \right) \mathbf{P} \mathbf{Y} - \frac{\gamma^3 (f'(\tau^\circ))^2}{n^2 (1 + \gamma f(\tau^\circ))} \boldsymbol{\psi} \boldsymbol{\psi}^\top \mathbf{P} \mathbf{Y} + O(n^{-5/2}) \text{ a.s.} \tag{G.7}$$

Define the two random variables in an arbitrary point $\mathbf{x}$: $\psi_{\mathbf{x}} = \|\boldsymbol{\epsilon}_{\mathbf{x}}\|^2 - \mathbb{E}\|\boldsymbol{\epsilon}_{\mathbf{x}}\|^2$ and $\boldsymbol{\epsilon}_{\mathbf{x}} = (\mathbf{x} - \mu)/\sqrt{p}$. Combining (G.7) and Corollary 2,

$$\begin{aligned} \hat{\alpha}^\top k(\mathbf{x}) &= \mathcal{B}_{(2,n^{-1})}^\top k_{(1,1)} + \mathcal{B}_{(2,n^{-3/2})}^\top k_{(1,1)} + \mathcal{B}_{(2,n^{-1})}^\top k_{(1,n^{-1/2})} + \mathcal{B}_{(2,n^{-2})}^\top k_{(1,1)} \\ &\quad + \mathcal{B}_{(2,n^{-3/2})}^\top k_{(1,n^{-1/2})} + \mathcal{B}_{(2,n^{-1})}^\top k_{(1,n^{-1})} + O(n^{-3/2}) \text{ a.s.} \\ &= \mathcal{B}_{(3,1)} + \mathcal{B}_{(3,n^{-1/2})} + \mathcal{B}_{(3,n^{-1})} + O(n^{-3/2}) \text{ a.s.} \end{aligned} \tag{G.8}$$

where $\mathcal{B}_{(3,1)} = 0$, $\mathcal{B}_{(3,n^{-1/2})} = \frac{\gamma f'(\tau^\circ)}{n} \boldsymbol{\psi}^\top \mathbf{P} \mathbf{Y} - \frac{2\gamma f'(\tau^\circ)}{n} \mathbf{Y}^\top \mathbf{P} \mathbf{W}^\top \boldsymbol{\epsilon}_x$ and $\mathcal{B}_{(3,n^{-1})} = \frac{\gamma f''(\tau^\circ)}{n} \mathbf{Y}^\top \mathbf{P} \boldsymbol{\psi} \psi_x + \frac{\gamma f''(\tau^\circ)}{2n} \mathbf{Y}^\top \mathbf{P} \boldsymbol{\psi}^2$. Finally, $\hat{g}(\mathbf{x}) = \hat{b} + \hat{\boldsymbol{\alpha}}^\top k(\mathbf{x})$ can be obtained from (G.6) and (G.8)

$$\begin{aligned} \hat{g}(\mathbf{x}) &= \hat{b} + \hat{\boldsymbol{\alpha}}^\top k(\mathbf{x}) \\ &= \mathcal{B}_{(1,1)} + \mathcal{B}_{(3,1)} + \mathcal{B}_{(1,n^{-1/2})} + \mathcal{B}_{(3,n^{-1/2})} + \mathcal{B}_{(1,n^{-1})} + \mathcal{B}_{(3,n^{-1})} + O(n^{-3/2}) \text{ a.s.} \\ &= \bar{Y} - \frac{2\gamma f'(\tau^\circ)}{n} \mathbf{Y}^\top \mathbf{P} \mathbf{W}^\top \boldsymbol{\epsilon}_x + \frac{\gamma f''(\tau^\circ)}{n} \mathbf{Y}^\top \mathbf{P} \boldsymbol{\psi} \psi_x + O(n^{-3/2}) \text{ a.s.} \end{aligned}$$

The above equation can be greatly simplified by noting that $\mathbf{Y}^\top \mathbf{P} = \mathbf{Y}^\top - \bar{Y} \mathbf{1}_n^\top$, $\mathbf{1}_n^\top \mathbf{W}^\top = 0$, by the strong law of large numbers $\mathbf{1}_n^\top \boldsymbol{\psi} \rightarrow 0$, a.s. and hence

$$\begin{aligned} \hat{g}(\mathbf{x}) &= \bar{Y} - \frac{2\gamma f'(\tau^\circ)}{n} \mathbf{Y}^\top \mathbf{W}^\top \boldsymbol{\epsilon}_x + \frac{\gamma f''(\tau^\circ)}{n} \mathbf{Y}^\top \boldsymbol{\psi} \psi_x + O(n^{-3/2}) \text{ a.s.} \\ &= \tilde{g}(\mathbf{x}) + O(n^{-3/2}) \text{ a.s.,} \end{aligned}$$

H. Proof of Theorem 7

The LS-SVR approximation can be written as a linear smoother

$$\begin{aligned} \tilde{g}(\mathbf{x}) &= \bar{Y} - \frac{2\gamma f'(\tau)}{n} \mathbf{Y}^\top \mathbf{W}^\top \boldsymbol{\epsilon}_x + \frac{\gamma f''(\tau)}{n} \mathbf{Y}^\top \boldsymbol{\psi} \psi_x \\ &= \sum_{i=1}^n \left\{ \frac{1}{n} - \frac{2\gamma f'(\tau)}{n} \sum_{j=1}^p (\epsilon_{ij} - \bar{\epsilon}_j) \epsilon_{x_j} + \frac{\gamma f''(\tau)}{n} [\psi_x \|\boldsymbol{\omega}_i\|^2 - \psi_x \mathbb{E} \|\boldsymbol{\omega}_i\|^2] \right\} Y_i \\ &= \sum_{i=1}^n l_i(\mathbf{x}) Y_i = \sum_{i=1}^n \xi_i \end{aligned}$$

where ϵ_{x_j} denotes the j th component of the vector $\boldsymbol{\epsilon}_x$. The conditional mean of $\tilde{g}(\mathbf{x})$ is

$$\mathbb{E} \{ \tilde{g}(\mathbf{x}) \mid \mathbb{X} \} = \mathbb{E} \left\{ \sum_{i=1}^n l_i(\mathbf{x}) Y_i \mid \mathbb{X} \right\} = \sum_{i=1}^n [\mathbf{w}^\top \varphi(\mathbf{x}_i) + b] l_i(\mathbf{x})$$

with $\mathbb{X}$ denoting the sequence $\{\mathbf{X}_1, \dots, \mathbf{X}_n, \dots\}$. Similarly, the conditional variance of $\tilde{g}(\mathbf{x})$ is

$$\mathbb{V}\text{ar} \{ \tilde{g}(\mathbf{x}) \mid \mathbb{X} \} = \mathbb{V}\text{ar} \left\{ \sum_{i=1}^n l_i(\mathbf{x}) Y_i \mid \mathbb{X} \right\} = \sigma_e^2 \sum_{i=1}^n l_i^2(\mathbf{x})$$

For the individual ξ_i we have $\mu_i = \mathbb{E} \{ \xi_i \mid \mathbb{X} \} = [\mathbf{w}^\top \varphi(\mathbf{x}_i) + b] l_i(\mathbf{x})$ and $\mathbb{V}\text{ar} \{ \xi_i \mid \mathbb{X} \} = \sigma_e^2 l_i^2(\mathbf{x})$. Since $Y_i = \mathbf{w}^\top \varphi(\mathbf{x}_i) + b + e_i$, we have

$$|\xi_i - \mathbb{E}(\xi_i \mid \mathbb{X})| = \left| l_i(\mathbf{x}) (Y_i - \mathbf{w}^\top \varphi(\mathbf{x}_i) - b) \right| = |l_i(\mathbf{x}) e_i|.$$

Let $D_n^2 = \sigma_e^2 \sum_{i=1}^n l_i^2(\mathbf{x}) = \sigma_e^2 \|l(\mathbf{x})\|^2$, for any $L > 0, \varepsilon > 0$ and by Chebyshev's inequality

$$\begin{aligned}
& \mathbb{E}[(\xi_i - \mu_i)^2 \mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n) \mid \mathbb{X}] \\
&= \mathbb{E}[l_i^2(\mathbf{x}) e_i^2 \{ \mathbf{I}(|e_i| \leq L) + \mathbf{I}(|e_i| > L) \} \mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n) \mid \mathbb{X}] \\
&\leq l_i^2(\mathbf{x}) L^2 \mathbb{E}[\mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n) \mid \mathbb{X}] + l_i^2(\mathbf{x}) \mathbb{E}[e_i^2 \mathbf{I}(|e_i| > L) \mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n) \mid \mathbb{X}] \\
&\leq l_i^2(\mathbf{x}) L^2 \frac{\mathbb{E}[|\xi_i - \mu_i|^2 \mid \mathbb{X}]}{\varepsilon^2 D_n^2} + l_i^2(\mathbf{x}) \mathbb{E}[e_i^2 \mathbf{I}(|e_i| > L) \mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n) \mid \mathbb{X}] \\
&= l_i^2(\mathbf{x}) L^2 \frac{\sigma_e^2 l_i^2(\mathbf{x})}{\varepsilon^2 D_n^2} + l_i^2(\mathbf{x}) \mathbb{E}[e_i^2 \mathbf{I}(|e_i| > L) \mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n) \mid \mathbb{X}]. \tag{H.1}
\end{aligned}$$

Using (H.1), we have

$$\begin{aligned}
& \frac{1}{D_n^2} \sum_{i=1}^n \mathbb{E}[(\xi_i - \mu_i)^2 \mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n \mid \mathbb{X})] \\
&\leq \frac{1}{D_n^2} \sum_{i=1}^n \left\{ l_i^2(\mathbf{x}) L^2 \frac{\sigma_e^2 l_i^2(\mathbf{x})}{\varepsilon^2 D_n^2} + l_i^2(\mathbf{x}) \mathbb{E}[e_i^2 \mathbf{I}(|e_i| > L) \mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n) \mid \mathbb{X}] \right\} \\
&\leq \frac{L^2}{\varepsilon^2 \sigma_e^2} \max_{1 \leq k \leq n} \frac{l_k^2(\mathbf{x})}{\|l(\mathbf{x})\|^2} + \frac{1}{\sigma_e^2} \max_{1 \leq k \leq n} \mathbb{E}[e_k^2 \mathbf{I}(|e_k| > L) \mathbf{I}(|\xi_k - \mu_k| \geq \varepsilon D_n) \mid \mathbb{X}].
\end{aligned}$$

By Corollary 1, it can be shown that the first term

$$\max_{1 \leq k \leq n} \frac{l_k^2(\mathbf{x})}{\|l(\mathbf{x})\|^2} = O_p\left(\frac{1}{n}\right).$$

Hence, by taking the limit for $n \rightarrow \infty$, the first term converges to zero in probability. By fixing L , Lindeberg's condition (Billingsley, 1995, Theorem 27.2) yields

$$\begin{aligned}
& \limsup_{n \rightarrow \infty} \frac{1}{D_n^2} \sum_{i=1}^n \mathbb{E}[(\xi_i - \mu_i)^2 \mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n \mid \mathbb{X})] \\
&\leq \limsup_{n \rightarrow \infty} \frac{1}{\sigma_e^2} \max_{1 \leq k \leq n} \mathbb{E}[e_k^2 \mathbf{I}(|e_k| > L) \mathbf{I}(|\xi_k - \mu_k| \geq \varepsilon D_n) \mid \mathbb{X}].
\end{aligned}$$

By letting $L \rightarrow \infty$, it follows that $\mathbf{I}(|e_k| > L) \rightarrow 0$, and since $\mathbb{E}[\mathbf{Y}^2 \mid \mathbb{X}] < \infty$,

$$\lim_{n \rightarrow \infty} \frac{1}{D_n^2} \sum_{i=1}^n \mathbb{E}[(\xi_i - \mu_i)^2 \mathbf{I}(|\xi_i - \mu_i| \geq \varepsilon D_n \mid \mathbb{X})] = 0.$$

I. More simulation results

In this section, we include more simulation results for the interested reader. Figures 6 to 7 and Figures 8 to 9 illustrate more results under model (5.1) and model (5.2) with varying covariance settings, bandwidths, and regularization parameters, respectively.

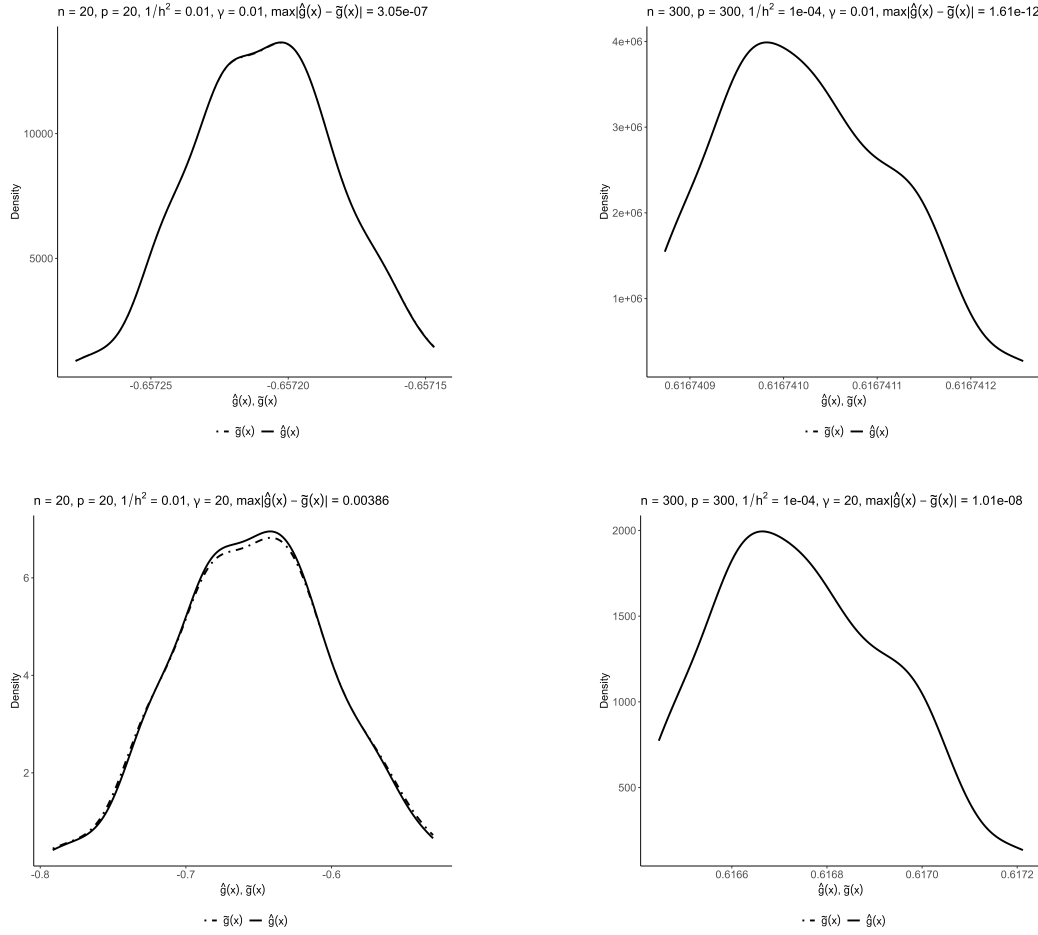


Figure 6: Kernel density estimates for $\hat{g}(\mathbf{x})$ (full) and $\tilde{g}(\mathbf{x})$ (dash dot) for varying n and p combinations under (5.1) with $\mathbf{\Omega} = \mathbf{I}_p$, $1/h^2 = (0.01, 0.0001)$ and $\gamma = (0.01, 20)$.

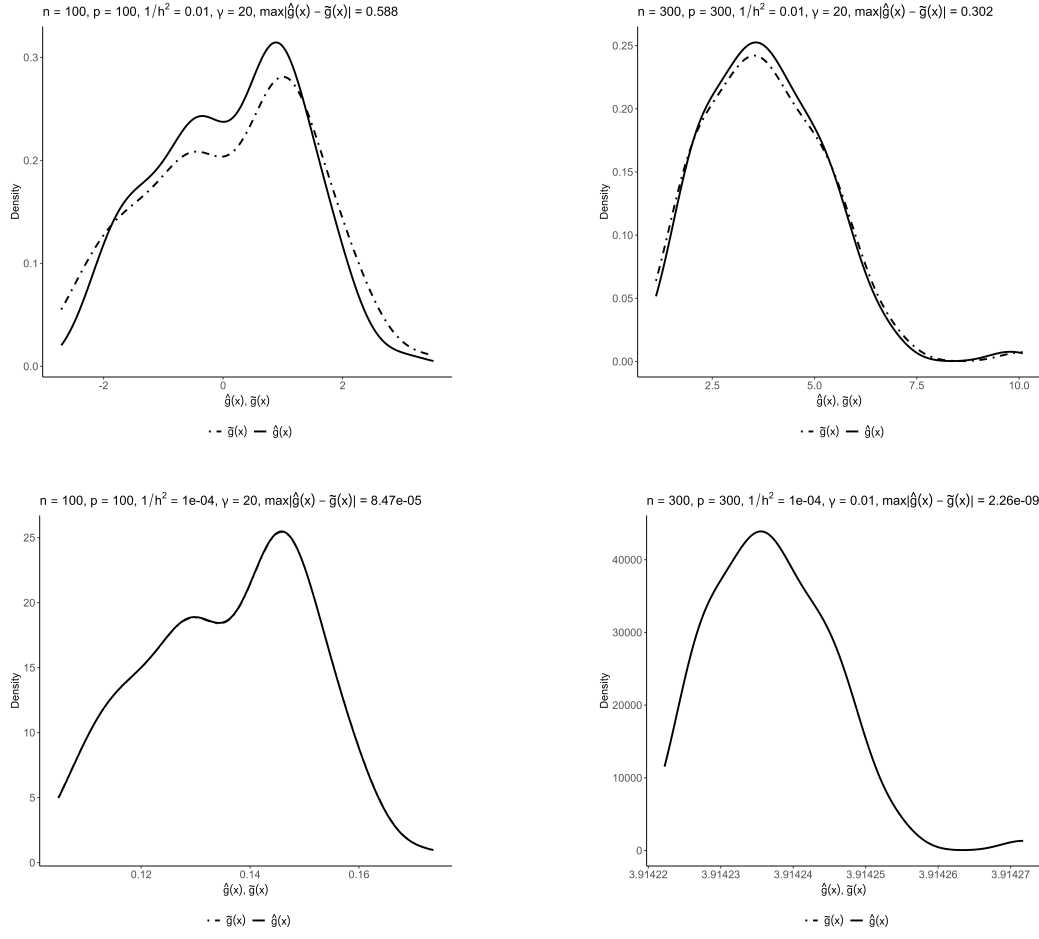


Figure 7: Kernel density estimates for $\hat{g}(\mathbf{x})$ (full) and $\tilde{g}(\mathbf{x})$ (dash dot) for varying n and p combinations under (5.1) with $\text{AR}_p(1)$ covariance structure, $1/h^2 = (0.01, 0.0001)$ and $\gamma = (0.01, 20)$.

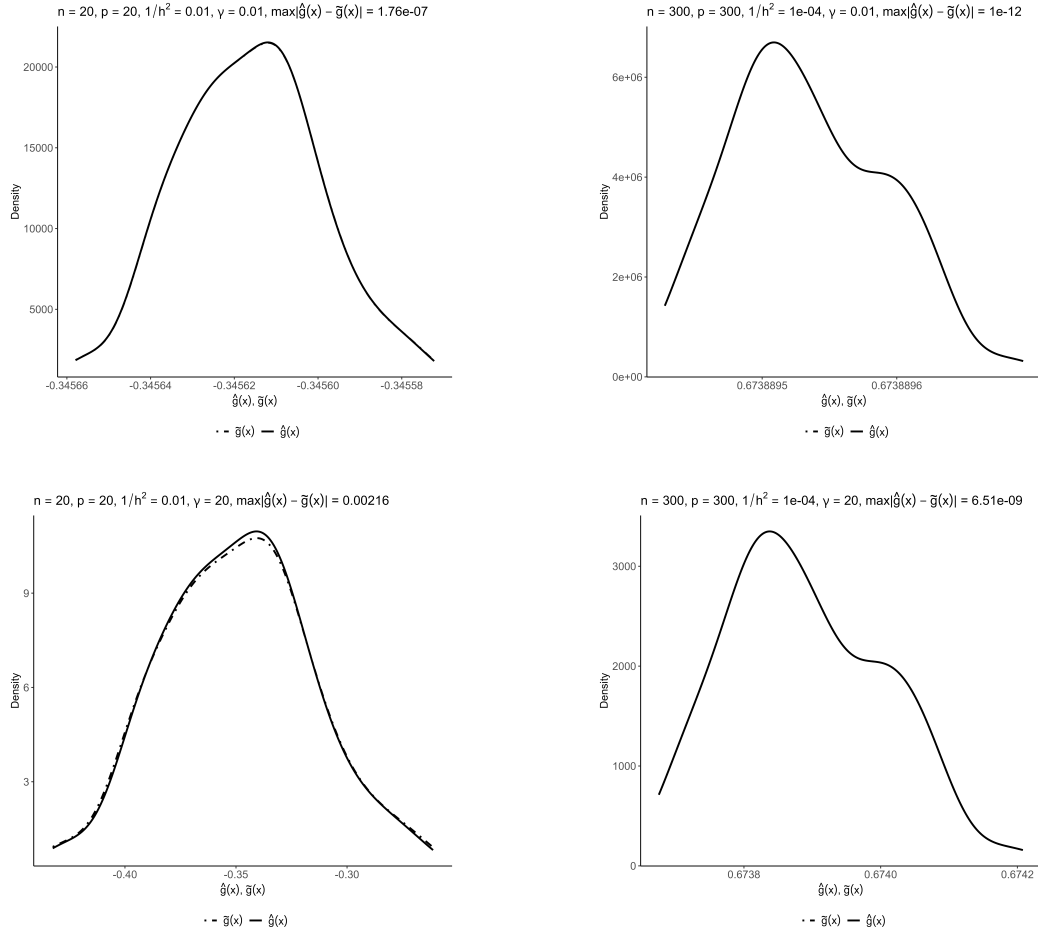


Figure 8: Kernel density estimates for $\hat{g}(\mathbf{x})$ (full) and $\tilde{g}(\mathbf{x})$ (dash dot) for varying n and p combinations under (5.2) with $\mathbf{\Omega} = \mathbf{I}_p$, $1/h^2 = (0.01, 0.0001)$ and $\gamma = (0.01, 20)$.

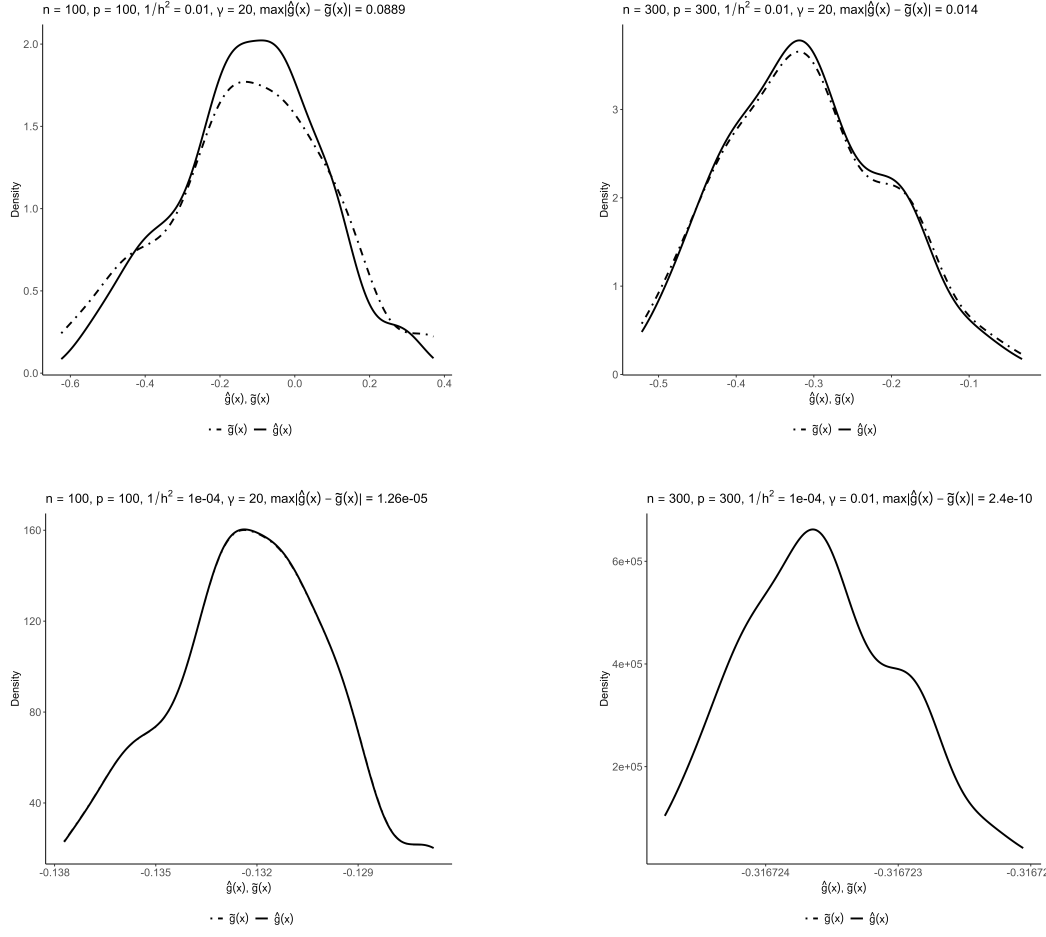


Figure 9: Kernel density estimates for $\hat{g}(\mathbf{x})$ (full) and $\tilde{g}(\mathbf{x})$ (dash dot) for varying n and p combinations under (5.2) with $\text{AR}_p(1)$ covariance structure, $1/h^2 = (0.01, 0.0001)$ and $\gamma = (0.01, 20)$.

References

- Z. D. Bai and J. W. Silverstein. *Spectral Analysis of Large Dimensional Random Matrices*. Springer, New York, 2010.
- F. Benaych-Georges and A. Knowles. Lectures on the local semicircle law for wigner matrices, 2018. URL <https://arxiv.org/abs/1601.04055>.
- F. Benaych-Georges and R. R. Nadakuditi. The eigenvalues and eigenvectors of finite, low rank perturbations of large random matrices. *Advances in Mathematics*, 227:494–521, 2011. doi: 10.1016/j.aim.2011.03.011. URL <https://arxiv.org/abs/0910.2120>.
- P. J. Bickel and E. Levina. Covariance regularization by thresholding. *Annals of Statistics*, 36(6):2577–2604, 2008a.
- P. J. Bickel and E. Levina. Regularized estimation of large covariance matrices. *Annals of Statistics*, 36(1):199–227, 2008b.
- P. Billingsley. *Probability and Measure*. John Wiley & Sons, New York, 3rd edition, 1995.
- T. T. Cai, C-H. Zhang, and H.H. Zhou. Optimal rates of convergence for covariance matrix estimation. *Annals of Statistics*, 38(4):2118–2144, 2010.
- A. Caponnetto and E. De Vito. Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics*, 7:331–368, 2007.
- C. Cortes and V. Vapnik. Support-vector networks. *Machine learning*, 20:273–297, 1995.
- R. Couillet and F. Benaych-Georges. Kernel spectral clustering of large dimensional data. *Electronic Journal of Statistics*, 10(1):1393–1454, 2016.
- R. Couillet, Z. Liao, and X. Mai. Classification asymptotics in the random matrix regime. In *2018 26th European Signal Processing Conference (EUSIPCO)*, pages 1875–1879. IEEE, 2018.
- K. De Brabanter, J. De Brabanter, J. A. K. Suykens, and B. De Moor. Optimized fixed-size kernel models for large data sets. *Computational Statistics & Data Analysis*, 54: 1484–1504, 2010.
- K. De Brabanter, J. De Brabanter, J. A. K. Suykens, and B. De Moor. Approximate confidence and prediction intervals for least squares support vector regression. *IEEE Transactions on Neural Networks*, 22(1):110–120, 2011.
- H. Drucker, C. J. Burges, L. Kaufman, A. Smola, and V. Vapnik. Support vector regression machines. *Advances in neural information processing systems*, 9, 1996.
- N. El Karoui. The spectrum of kernel random matrices. *The Annals of Statistics*, 38(1): 1–50, 2010.
- J. Fan, Y. Liao, and M. Mincheva. Large covariance estimation by thresholding principal orthogonal complements. *Journal of the Royal Statistical Society, Series B*, 75(4):603–680, 2013.

- J. Friedman, T. Hastie, and R. Tibshirani. Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, 9(3):432–441, 2008.
- D. L. Hanson and F. T. Wright. A bound on tail probabilities for quadratic forms in independent random variables. *The Annals of Mathematical Statistics*, 42(3):1079–1083, 1971.
- R. A. Horn and C. R. Johnson. *Matrix Analysis*. Matrix Analysis. Cambridge University Press, 2013. ISBN 9780521839402.
- G-B. Huang, H. Zhou, X. Ding, and R. Zhang. Extreme learning machine for regression and multiclass classification. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, 42(2):513–529, 2011.
- Y. Igarashi, N. Nakatsu, T. Yamashita, A. Ono, Y. Ohno, T. Urushidani, and H. Yamada. Open tg-gates: a large-scale toxicogenomics database. *Nucleic acids research*, 43(D1):D921–D927, 2015.
- I. M. Johnstone. On the distribution of the largest eigenvalue in principal components analysis. *Annals of Statistics*, 29(2):295–327, 2001.
- A. Kammoun and M-S. Alouini. On the precise error analysis of support vector machines. *IEEE Open Journal of Signal Processing*, 2:99–118, 2021.
- S. Kim, J. Chen, T. Cheng, A. Gindulyte, J. He, S. He, Q. Li, B. A. Shoemaker, P. A. Thiessen, and Bo B. Yu. Pubchem in 2021: new data content and improved web interfaces. *Nucleic acids research*, 49(D1):D1388–D1395, 2021.
- O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.
- Z. Liao and R. Couillet. A large dimensional analysis of least squares support vector machines. *IEEE Transactions on Signal Processing*, 67(4):1065–1074, 2019.
- J. Mercer. Functions of positive and negative type, and their connection the theory of integral equations. *Philosophical transactions of the royal society of London. Series A, containing papers of a mathematical or physical character*, 209(441-458):415–446, 1909.
- F. Merlevède, M. Peligrad, and E. Rio. Bernstein inequality and moderate deviations under strong mixing conditions. *The Annals of Probability*, 37(4):1427–1459, 2009. doi: 10.1214/08-AOP435.
- H. Moriwaki, Y-S. Tian, N. Kawashita, and T. Takagi. Mordred: a molecular descriptor calculator. *Journal of cheminformatics*, 10(1):4, 2018.
- M. Pourahmadi. *High-Dimensional Covariance Estimation: With High-Dimensional Data*. John Wiley & Sons, Hoboken, NJ, 2013.
- N. Puchkin and M. Rakhuba. Dimension-free structured covariance estimation. In *Proceedings of the 37th Conference on Learning Theory*, Proceedings of Machine Learning Research, 2024.

- C. Doz C. Ren, J-P. Ovarlez, and R. Couillet. Large dimensional analysis of ls-svm transfer learning: Application to polsar classification. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE, 2023.
- P. Rigollet and J.-C. Hütter. High dimensional statistics. <https://arxiv.org/abs/2310.19244>, 2015.
- M. Rudelson and R. Vershynin. Hanson-wright inequality and sub-gaussian concentration. *Electronic Communications in Probability*, 18, 2013.
- W. Rudin. *Functional Analysis*. McGraw–Hill, 2 edition, 1991.
- B. Schölkopf and A. J. Smola. *Learning with kernels: support vector machines, regularization, optimization, and beyond*. MIT press, 2002.
- J. Shawe-Taylor and N. Cristianini. *Kernel methods for pattern analysis*. Cambridge University Press, 2004.
- A. J. Smola and B Schölkopf. A tutorial on support vector regression. *Statistics and computing*, 14:199–222, 2004.
- I. Steinwart, D. R. Hush, and C. Scovel. Optimal rates for regularized least squares regression. In *COLT*, pages 79–93, 2009.
- J. A. K. Suykens and J. Vandewalle. Least squares support vector machine classifiers. *Neural processing letters*, 9:293–300, 1999.
- J. A. K. Suykens, T. Van Gestel, J. De Brabanter, B. De Moor, and J. Vandewalle. *Least Squares Support Vector Machines*. World Scientific, Singapore, 2002. ISBN 9789812381514.
- M. Swain. Pubchempy: Python wrapper for the pubchem pug rest api. <https://pubchempy.readthedocs.io>, 2014.
- T. Van Gestel, J. A. K. Suykens, G. Lanckriet, A. Lambrechts, B. De Moor, and J. Vandewalle. Bayesian framework for least-squares support vector machine classifiers, gaussian processes, and kernel fisher discriminant analysis. *Neural computation*, 14(5):1115–1147, 2002.
- V. Vapnik. *The nature of statistical learning theory*. Springer, 1999.
- R. Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2018.
- M. J. Wainwright. *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, volume 48 of *Cambridge Series in Statistical and Probabilistic Mathematics*. Cambridge University Press, Cambridge, UK, 2019.
- L. Wang, T. Zhang, and R. Li. High-dimensional covariance estimation via constrained l_q -type regularization. *Mathematics*, 11(4):1022, 2023.

- J. Ye and T. Xiong. Svm versus least squares svm. In *Artificial intelligence and statistics*, pages 644–651. PMLR, 2007.
- Y. Zhang, S. Huang, and Z. Chen. Estimation of large-dimensional covariance matrices via second-order stein-type regularization. *Entropy*, 25(1):53, 2023.